

Contents

1	Mathematical Models of Systems	1
1.1	Exercises	1
1.2	Advanced problems	6
1.3	State space models	11
2	System Analysis	17
2.1	The performance of feedback control systems	17
2.2	The stability of linear systems	28
3	Root Locus	33
3.1	Root locus analysis	33
4	Analysis in Frequency Domain	39
4.1	Bode diagrams	39
4.2	Nyquist diagrams	42
5	Control System Design	45
5.1	Problems	45
A	Introduction to Matlab	51
A.1	Introduction	51
A.2	Statements and variables	51
A.3	Entering vectors and matrices	52
A.4	Functions	54
A.5	Polynomials	55
A.6	Loops and logical statements	56
A.7	Plotting	56
A.8	Toolboxes and m-files	56
B	Mathematical background	61
B.1	Linear approximation	61
B.2	Laplace transform	61
C	Analysis of first- and second-order systems	63
C.1	Step response of a first-order system	63
C.2	Step response of a second-order system	64

D Root Locus	67
Bibliography	69

Chapter 1

Mathematical Models of Systems

1.1 Exercises

Ex. 1 A nonlinear device is represented by the function

$$y = f(x) = x^{1/2}$$

where the operating point for the input x is $x_0 = 1/2$. Determine the linear approximation of the function.

Ex. 2 The output y and the input x of a device are related by:

$$y = x + 0.4x^3$$

1. Find the values of the output for steady-state operation at two operating points: $x_0 = 1$ and $x_0 = 2$.
2. Obtain a linearized model for both operating points and compare them.

Ex. 3 The transfer function of a system is:

$$\frac{Y(s)}{U(s)} = \frac{10(s+2)}{s^2 + 8s + 15}$$

Determine $y(t)$ when $u(t)$ is a unit step input.

Ex. 4 Obtain the unit impulse response and the unit step response of a unity feedback system whose open-loop transfer function is

$$G(s) = \frac{2s+1}{s^2}$$

Plot the results.

Ex. 5 An input

$$r(t) = e^{-t}$$

is applied to a black box with a transfer function $G(s)$. The resulting output, when the initial conditions are zero, is:

$$c(t) = 2 - 3e^{-t} + 3e^{-2t}\cos 2t, \quad t \geq 0$$

Determine $G(s)$ for this system.

Ex. 6 A system is represented in Figure 1.1.

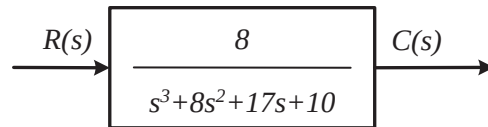


Figure 1.1:

1. Determine the partial fraction expansion and $c(t)$ for a ramp input, $r(t) = t$, $t \geq 0$.
2. Obtain a plot of $c(t)$ for part (1), and find $c(t)$ for $t = 1.5$ seconds.
3. Determine the impulse response of the system $c(t)$ for $t \geq 0$.
4. Obtain a plot of $c(t)$ for part (3) and find $c(t)$ for $t = 1.5$ seconds.

Ex. 7 Figure 1.2 shows a position control system.

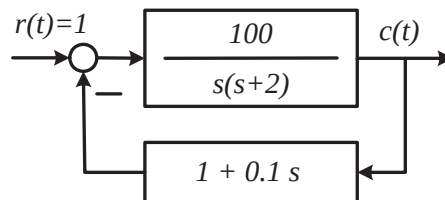


Figure 1.2:

What is the response $c(t)$ to a unit step input? Use Matlab to plot and check the results.

Ex. 8 A system is shown in Figure 1.3.

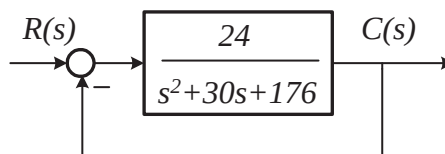


Figure 1.3:

1. Find the closed-loop transfer function $C(s)/R(s)$.
2. Determine $C(s)$ when the input $r(t)$ is a unit step.
3. Find $c(t)$ and plot it, also showing the step input.

Ex. 9 A closed-loop control system is shown in Figure 1.4.

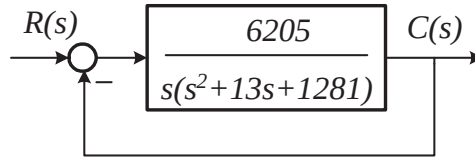


Figure 1.4:

1. Determine the transfer function $T(s) = C(s)/R(s)$.
2. Determine the poles and zeros of $T(s)$.
3. Use a unit step $R(s) = 1/s$ and obtain the partial fraction expansion for $C(s)$ and the value of the residues.
4. Plot $c(t)$ and discuss the effect of the real and complex poles of $T(s)$. Do the complex poles or the real poles dominate the response?

Ex. 10 A closed-loop control system is shown in Figure 1.5.

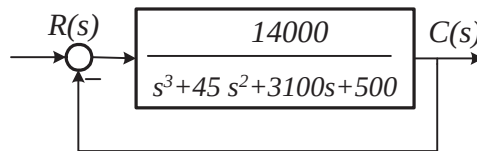


Figure 1.5:

1. Determine the transfer function $T(s) = C(s)/R(s)$.
2. Determine the poles and zeros of $T(s)$.
3. Use a unit step $R(s) = 1/s$ and obtain the partial fraction expansion for $C(s)$ and the value of the residues.
4. Plot $c(t)$ and discuss the effect of the real and complex poles of $T(s)$. Do the complex poles or the real poles dominate the response?
5. Predict the final value of $c(t)$ for the unit step input.

Ex. 11 The rotational velocity ω of the satellite shown in Figure 1.6 is adjusted by changing the length of the beam L . The transfer function between $\omega(s)$ and the incremental change in beam length is

$$\frac{\omega(s)}{\Delta L(s)} = \frac{2.5(s+2)}{(s+5)(s+1)^2}$$

The beam length change is $\Delta L(s) = 1/(4s)$. Determine the response of the velocity $\omega(t)$.

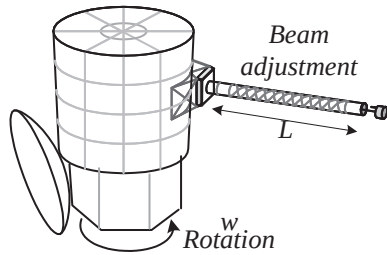


Figure 1.6: Satellite

Ex. 12 Determine the overall transfer function of the systems shown in Figures 1.7, 1.8, 1.9, 1.10:

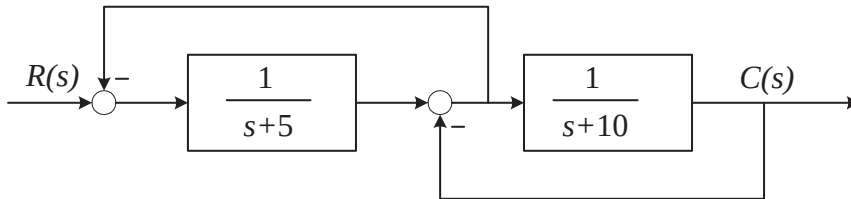


Figure 1.7:

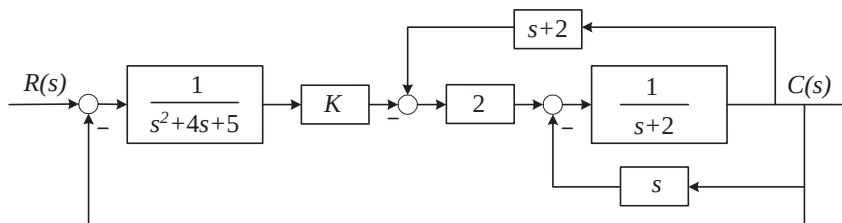


Figure 1.8:

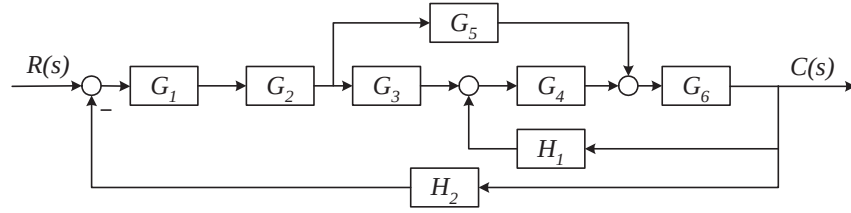


Figure 1.9:

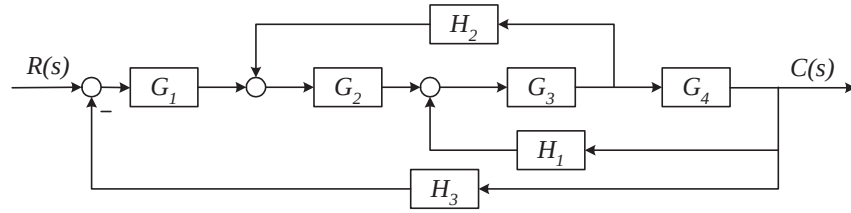


Figure 1.10:

Ex. 13 A system is described by the following equations:

$$\begin{aligned}
 E(s) &= R(s) - C(s) \cdot H_3(s) \\
 U_1(s) &= E(s) \cdot G_1(s) \\
 U_3(s) &= [U_1(s) - U_2(s)] \cdot G_2(s) \\
 U_2(s) &= U_4(s) \cdot H_2(s) \\
 U_4(s) &= [U_3(s) + U_5(s)] \cdot G_3(s) \\
 C(s) &= U_4(s) \cdot G_4(s) \\
 U_5(s) &= C(s) \cdot H_1(s)
 \end{aligned}$$

where $H_1(s)$, $H_2(s)$, $H_3(s)$, $G_1(s)$, $G_2(s)$, $G_3(s)$, $G_4(s)$ are transfer functions and E , R , C , U_1 , U_2 , U_3 , U_4 are signals.

1. Obtain the block diagram of the system and simplify it.
2. Obtain the overall transfer function $M(s) = C(s)/R(s)$.

Ex. 14 Consider the following ordinary differential equation system:

$$\begin{aligned}
 \frac{dx_1}{dt} &= x_2 \\
 \frac{dx_2}{dt} &= \frac{1}{M}[f(t) - Bx_2 - Kx_1]
 \end{aligned}$$

with the initial conditions $x_{10} = 0$; $x_{20} = 0$, where $f(t) = 25$; $M = 1$; $B = 5$; $K = 25$.

Use the *ode23* function to solve the system over a time interval of 0 to 3, and plot in the same graphic window (use *subplot* function): one plot of x_1 and x_2 versus time; and one plot of x_2 versus x_1 . Use titles and labels on the plots.

1.2 Advanced problems

P. 1 Continuous system. Consider a continuous time system with a true model given by

$$\frac{dx(t)}{dt} = f(x(t), u(t)) = -\sqrt{x(t)} + \frac{u(t)^2}{3}$$

Assume that the input $u(t)$ fluctuates around $u_0=2$. Find an operating point with $u_0=2$ and show that the linearized model around is:

$$x_0 = \frac{16}{9}, \quad \frac{d\Delta x(t)}{dt} = -\frac{3}{8}\Delta x(t) + \frac{4}{3}\Delta u(t)$$

with

$$\Delta u = u(t) - u_0, \quad \Delta x = x(t) - x_0$$

Consider $u(t)$ a square wave of increasing amplitude, as shown in Figure 1.11. Simulate

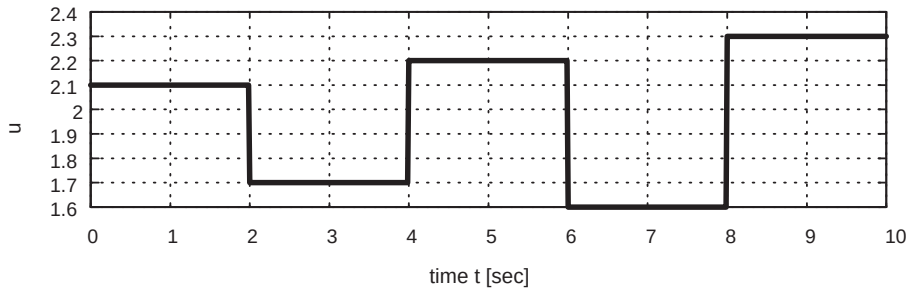


Figure 1.11: Function $u(t)$

the system for $0 < t < 10$ taking $u(t)$ as the input, in the case of nonlinear model and linear approximation of the model, and compare the results.

P. 2 Nuclear Reactor. The radiation level l , of a nuclear reactor is controlled by means of the position of control bars. The control system compares the setpoint, that is the desired radiation level r_r with the signal received from a radiation sensor r_c and an error signal ϵ is obtained. The error is examined by an electronic device which produces a voltage v , transmitted to an assembly motor-transmission which is able to alter the position of the control bars.

The observations made showed that the level of radiation, l , depends on the position of the control bars according to the relation:

$$l(t) = \dot{p}(t) + 4p(t) + 5.$$

On the other hand, the radiation sensor, after a sudden increase with 2 units of the radiation level, responds according to the equation:

$$r_c(t) = 6(1 - e^{-3t})$$

The electronic device which manipulates the error behaves as a second-order system with the transfer function:

$$G_d(s) = \frac{4}{s^2 + \frac{\sqrt{2}}{2}s + 4}$$

The transfer function of motor - transmission is:

$$\frac{P(s)}{V(s)} = \frac{K}{s + 4}$$

1. Obtain the transfer function of the radiation sensor and the transfer function from the radiation sensor to the position of control bars, $G_p(s)$. Draw the block diagram of the system and obtain the transfer function of the entire system $M(s) = R_r(s)/L(s)$.
2. Which is the maximum level of radiation for $K = 0.15$, when $r_r(t)$ is a unit step?

P. 3 Spring-Mass-Damper System. A spring-mass-damper system is shown in Figure 1.12.

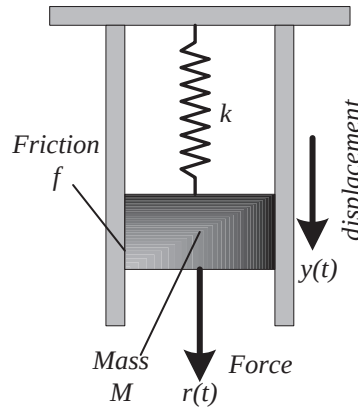


Figure 1.12: Spring-Mass-Damper System

The motion of the mass, denoted by $y(t)$ is described by Newton's second law of motion:

$$M \frac{d^2 y(t)}{dt^2} + f \frac{dy(t)}{dt} + ky(t) = r(t)$$

Determine the unforced response ($r(t) = 0$) of the spring-mass-damper system when the initial displacement is y_0 .

Plot $y(t)$ for the cases:

- 1) $y_0 = 0.15 \text{ m}$, $\frac{k}{M} = 2$, $\frac{f}{M} = 3$
- 2) $y_0 = 0.15 \text{ m}$, $\frac{k}{M} = 2$, $\frac{f}{M} = 1$
- 3) $y_0 = 0.15 \text{ m}$, $\frac{k}{M} = 2$, $f = 0$

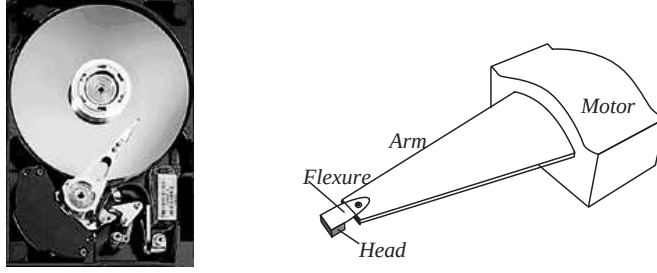


Figure 1.13: HardDisk Drive

P. 4 Hard Disk Control System. A disk drive read-write positioning system is shown in Figure 1.13.

The typical parameters for the disk drive are given in Table 1.1.

Table 1.1: Disk drive parameters

Parameter	Symbol	Typical Value
Inertia of arm and read head	J	$1 \text{ Nms}^2/\text{rad}$
Friction	b	20 kg/m/s
Amplifier	K_a	10-1000
Armature resistance	R	1Ω
Motor constant	K_m	5 Nm/A
Armature inductance	L	10^{-3} H

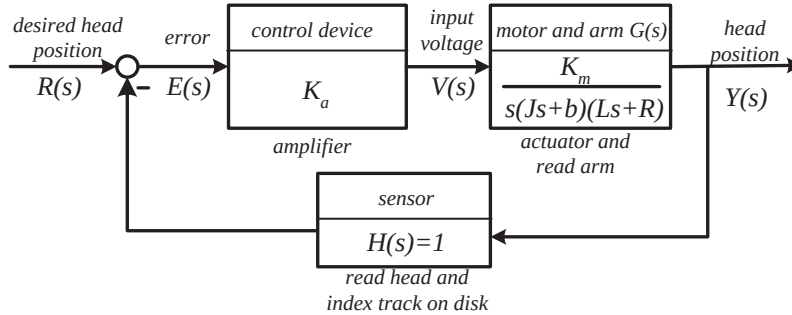


Figure 1.14: Hard Disk Control System

Using the block diagram from Figure 1.14 for the disk drive control system simulate the system unit step response for $K_a = 10, 100, 1000$ during one second and compare the results.

P. 5 Conic tank. Consider the conic tank shown in Figure 1.15. The volume of the liquid in the tank is equal to $kh(t)^3$, where k is a constant and h , the height of the liquid. The liquid is filled in the tank at a flow-rate q_e and discharged at a flow-rate q_s .

1. Obtain the equations which describe this system.

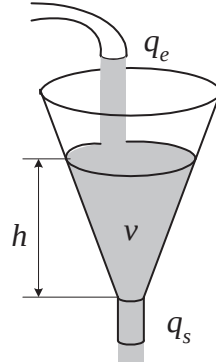


Figure 1.15: Conic tank

2. Linearize the equations around the equilibrium point $q_{e0} = 1$.
3. Draw the block diagram taking q_e as input and q_s as the output.
4. Determine the transfer function between q_e and q_s .
5. Calculate the final value of q_s , when q_e is a unit step.

Hint. The variables for this system are: the volume of liquid $v(t)$, the height of the liquid $h(t)$, the input flow-rate $q_e(t)$ and the output flow-rate $q_s(t)$. In order to obtain an input-output relation between q_e and q_s three equations are necessary. From the statement of the problem, the relation between the volume and the height is:

$$v(t) = kh(t)^3$$

The other two equations have to be found in the physical laws which govern the system. The volume of liquid in the tank is increased because of the difference between the input and output flow-rates, so:

$$q_e(t) - q_s(t) = \frac{dv(t)}{dt}$$

The third equation is a relation between the height of the liquid and the output flow-rate obtained from Toricelli's theorem in ideal conditions:

$$q_s(t) = CS\sqrt{2gh(t)} = B\sqrt{h(t)}$$

where C is the contraction coefficient and S the area of the outlet[m²] and $B = CS\sqrt{2g}$ is a constant.

P. 6 Magnetic Levitation. Magnetic levitation (MagLev) trains provide a high speed and a very low friction. The train floats on an air gap of about 1 cm as shown in Figure 1.16. Currently, maglev trains have been created in Germany and Japan for test runs only. There are two main methods for levitating maglev vehicles. The first method uses the power of magnetic "attraction" and the second is electrodynamic suspension using superconductor coils for magnetic "repulsion".

The levitation force F_L is controlled by the coil current i in the levitation coils and may be approximated by:

$$F_L = k \cdot \frac{i^2(t)}{z^2(t)}$$

where z is the air gap. This force is opposed by the downward force $F = mg$.

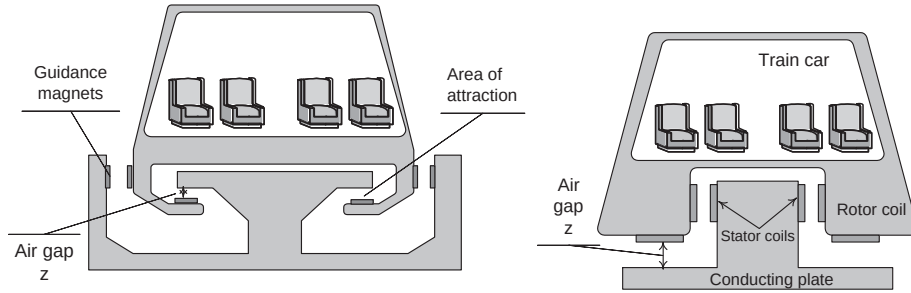


Figure 1.16: Maglev trains. a) Built based on attraction; b) Built based on repulsion

1. Determine the linearized relationship between the air gap z and the controlling current i near the equilibrium condition in the two cases.
2. Determine the transfer function from the input current $i(t)$ to the output position $z(t)$.
3. Plot the impulse response of the open loop systems for a period of time of 0.1 seconds (case a)) and 1 second (case b)).
4. The electrical system (levitating coil) can be represented by a voltage in series with an inductance and a resistance. The relationship between the input voltage u and the output current of the coil is $L di/dt + Ri = u$. Consider the control variable voltage u and obtain an overall transfer function for the maglev train positioning system.
5. Plot the impulse response for the system obtained at (4).
6. Obtain a state model of the form:

$$\dot{x} = f(x, u), \quad y = Cx$$

for the nonlinear system, and a state model of the form:

$$\dot{x} = Ax + Bu; \quad y = Cx$$

for the linearized system, where

$$x = [i \ z \ \dot{z}]^T; \quad C = [0 \ 1 \ 0]$$

Use the following values for simulation:

- Operating airgap $z_0 = 0.01$ m
- Mass of the train $m = 10000$ kg
- Levitation force constant $k = 0.001$ Nm²/A²
- Resistance and inductance $R = 0.3\Omega$, $L = 0.01H$

1.3 State space models

P. 1 A robot-arm drive system for one joint can be represented by the differential equation:

$$\frac{dv(t)}{dt} = -k_1v(t) - k_2y(t) + k_3i(t)$$

where $v(t)$ = velocity, $y(t)$ = position and $i(t)$ is the control-motor current. Put the equation in state variable form and set up the matrix form for $k_1 = k_2 = 1$.

P. 2 Determine the state variable description in the phase variable format for the system described by the transfer function:

$$\frac{C(s)}{R(s)} = \frac{4(s+3)}{(s+2)(s+4)}$$

P. 3 A system is described by its transfer function:

$$\frac{C(s)}{R(s)} = \frac{8(s+5)}{s^3 + 12s^2 + 44s + 48}$$

1. Determine the phase variable representation.
 2. Determine the canonical variable representation.
 3. Determine $\Phi(t)$ the state transition matrix.
- P. 4** Consider the spring-mass-damper system from problem **P. 3**/section 1.2.
1. Determine a state-space representation of the system.
 2. Plot the displacement of the mass and the mass velocity versus time.

For simulation consider the input force $r(t) = 0$, the initial displacement of the mass $y(0) = 0.15$ m, the initial velocity $\dot{y}(0) = 0$, $k/M = 2$, $f/M = 1$.

P. 5 A closed-loop control system is shown in Figure 1.17. Determine a state-space representation of the system.

P. 6 A closed-loop control system is shown in Figure 1.18.

1. Determine the closed-loop transfer function $T(s) = C(s)/R(s)$

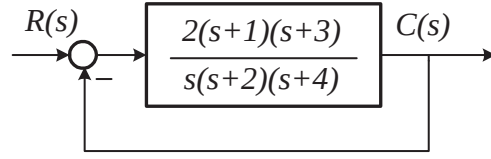


Figure 1.17: Closed-loop control system

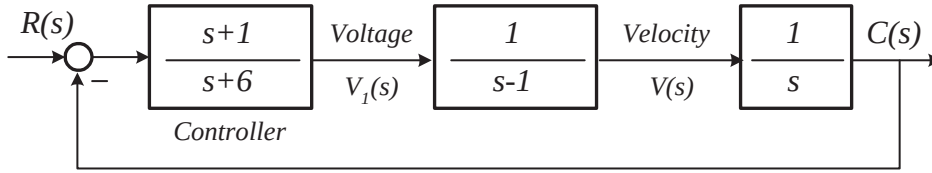


Figure 1.18: Closed-loop control system

2. Determine the matrix differential equation for the phase variable format.

P. 7 Consider a closed-loop control system shown in Figure 1.19.

1. Determine a state variable representation of the controller using Matlab.
2. Repeat part (1) for the plant.
3. With the controller and plant in state variable form, use Matlab to compute a closed-loop representation in state variable form and plot the closed-loop system impulse response.

P. 8 The transfer function of an electrical circuit is given by:

$$G(s) = \frac{1}{R_1 R_2 C_1 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_1 C_2)s + 1}$$

Let $R_1 = 1$; $R_2 = 10$; $C_1 = 0.5$ and $C_2 = 0.1$.

1. Show that a state variable representation is given by:

$$\dot{\mathbf{x}} = \begin{bmatrix} -3.2 & -2 \\ 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \mathbf{i}$$

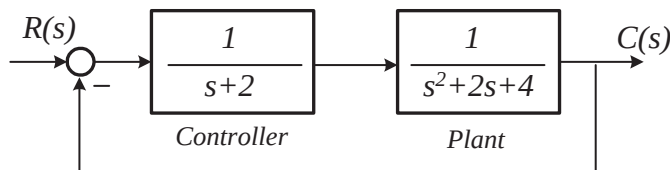


Figure 1.19: Closed-loop control system

- Using the state variable representation from part (1), plot the unit step response with the *step* function.

P. 9 Consider the system:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -2 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} \mathbf{u}; \quad \mathbf{y} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}$$

- Using the *ss2tf* function, determine the transfer function $Y(s)/U(s)$.
- Plot the response of the system to the initial condition $\mathbf{x}(0) = [0 \ 0 \ 1]^T$, for $0 \leq t \leq 10$, using the *initial* function.
- Compute the state transition matrix using the *expm* function, and determine $\mathbf{x}(t)$ at $t = 10$ for the initial condition given in part (2). Compare the result with the system response obtained in part (2).

P. 10 Consider the two systems:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -5 & -8 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix} \mathbf{u}; \quad \mathbf{y} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}$$

and

$$\dot{\mathbf{x}} = \begin{bmatrix} 0.5000 & 0.5000 & 0.7071 \\ -0.5000 & -0.5000 & 0.7071 \\ -6.3640 & -0.7071 & -8.0000 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix} \mathbf{u};$$

$$\mathbf{y} = \begin{bmatrix} 0.7071 & -0.7071 & 0 \end{bmatrix} \mathbf{x}$$

- Using the *ss2tf* function, determine the transfer function $Y(s)/U(s)$ for the first system.
- Repeat part (1) for the second system.
- Compare the results in parts (1) and (2) and comment.

P. 11 Consider the RLC circuit shown in Figure 1.20.

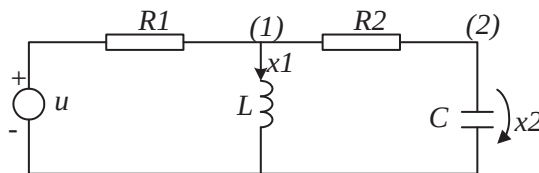


Figure 1.20: RLC Circuit

Natural state variables are currents through Ls and voltage drops across Cs. We can use either mesh or node analysis. Determine a state-variable model via node analysis, using the equations:

$$(1) : \frac{u - L\dot{x}_1}{R_1} = x_1 + \frac{L\dot{x}_1 - x_2}{R_2}$$

$$(2) : \frac{L\dot{x}_1 - x_2}{R_2} = C\dot{x}_2$$

P. 12 Consider a toy train consisting of an engine and a car. Assuming that the train only travels in one direction, we want to apply control to the train so that it has a smooth start-up and stop, along with a constant-speed ride.



Figure 1.21: Train system

The mass of the engine and the car will be represented by M_1 and M_2 , respectively. The two are held together by a spring, which has the stiffness coefficient of k . F represents the force applied by the engine, and the Greek letter, μ , represents the coefficient of rolling friction.

The system can be represented by the free body diagram shown in Figure 1.22.

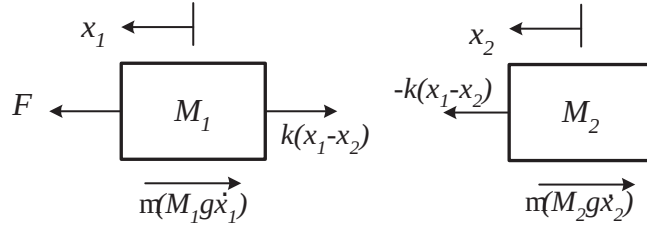


Figure 1.22: Free body diagram for the train system

From Newton's law, the sum of forces acting on a mass equals the mass times its acceleration. In this case, the forces acting on M_1 are the spring, the friction and the force applied by the engine. The forces acting on M_2 are the spring and the friction. In the vertical direction, the gravitational force is canceled by the normal force applied by the ground, so that there will be no acceleration in the vertical direction. The equations of motion in the horizontal direction are the followings:

$$M_1\ddot{x}_1 = F - k(x_1 - x_2) - \mu M_1 g \dot{x}_1$$

$$M_2\ddot{x}_2 = k(x_1 - x_2) - \mu M_2 g \dot{x}_2$$

1. Knowing that the state-variables are x_1 and x_2 , the output of the system is the velocity of the engine and the input is F , determine the state-variable equations and the matrices A , B , C , D from the general form of the state-space model:

$$\dot{\mathbf{x}} = \mathbf{A} \cdot \mathbf{x} + \mathbf{B} \cdot \mathbf{u}$$

$$\mathbf{y} = \mathbf{C} \cdot \mathbf{x} + \mathbf{D} \cdot \mathbf{u}$$

2. Using the time-domain equations, derive the transfer function $Y(s)/F(s)$. When finding the transfer function, zero initial conditions must be assumed.
3. Using Matlab function *tf2ss* determine the matrices A , B , C and D of the system. Compare them with the matrices obtained from the time-domain equations and comment.
4. Plot the step response using both the transfer function model and the state-space model.

Matlab representation

The numerical values of the variables are:

$$M_1 = 1 \text{ kg}$$

$$M_2 = 0.5 \text{ kg}$$

$$k = 1 \text{ N/sec}$$

$$F = 1 \text{ N}$$

$$\mu = 0.002 \text{ sec/m}$$

$$g = 9.8 \text{ m/s}^2$$

Chapter 2

System Analysis

2.1 The performance of feedback control systems

P. 1 A working knowledge of the relationship between the pole locations of the second-order closed-loop system with the transfer function

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

and the transient response is important in control design. Consider the four cases:

1. $\omega_n = 2, \zeta = 0$
2. $\omega_n = 2, \zeta = 0.1$
3. $\omega_n = 1, \zeta = 0$
4. $\omega_n = 1, \zeta = 0.2$

Plot the impulse response in each case, compare and discuss the result.

P. 2 Consider a system whose closed-loop poles and closed-loop zero are located in the s-plane on a line parallel to the $j\omega$ axis as shown in Figure 2.1. Show that the impulse response of such a system is a damped cosine function.

P. 3 Consider the system shown in Figure 2.2. The system is subjected to two signals, one the setpoint and the other the external disturbance. Show that the characteristic equation of this system (which contains information necessary to determine the basic characteristics of the system response) is the same regardless of which signal is chosen as input.

P. 4 A thermometer requires 1 minute to indicate 98% of the response to a step input. Assuming the thermometer is a first-order system, find the time constant.

P. 5 Consider the plots in Figure 2.3 representing the unit step responses of six systems with constant values at the numerators. Match the step response plots to the following characteristic equations:

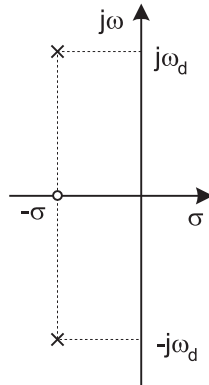


Figure 2.1: Closed-loop pole-zero configuration

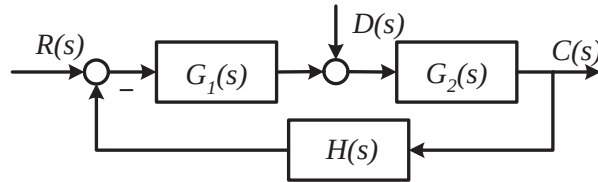


Figure 2.2: Closed-loop system with disturbance

1. $s + \frac{1}{2}$
2. $s + 2$
3. $s^2 + 1$
4. $s^2 + 9$
5. $s^2 + s + 9$
6. $s^2 + 2s + 9$

P. 6 Consider the unit step response of a unity-feedback control system whose open-loop transfer function is:

$$G(s) = \frac{1}{s(s+1)}$$

Obtain the rise time, peak time, maximum overshoot and settling time.

P. 7 The construction of the tunnel under the English Channel from France to Great Britain began in December 1987. The first connection of the boring tunnels from each country was achieved in November 1990. The machines operating from both ends of the channel bored toward the middle. To link up in the middle of the channel with necessary accuracy, a laser guidance system keep the machines precisely aligned. A model of the

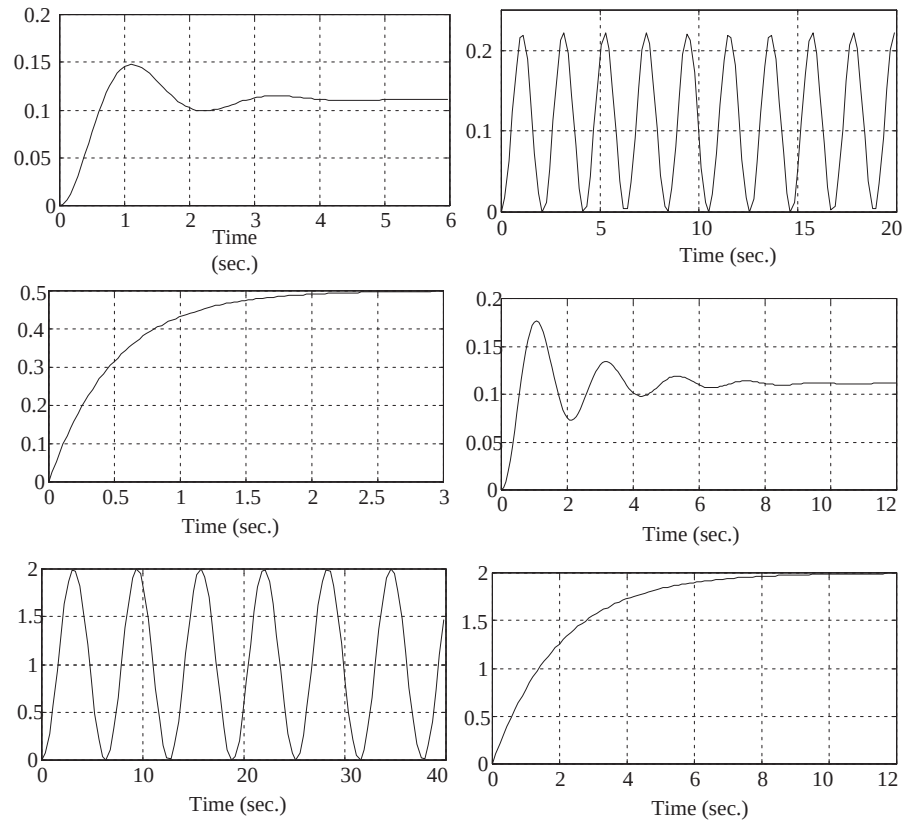


Figure 2.3: Step responses

boring machine is shown in Figure 2.4, when $C(s)$ is the actual angle of direction of travel of the boring machine and $R(s)$ is the desired angle. The effect of load on the machine is represented by disturbance $D(s)$. The design objective is to select the gain k so that the response to input changes is desirable while maintaining minimal error due to the disturbance.

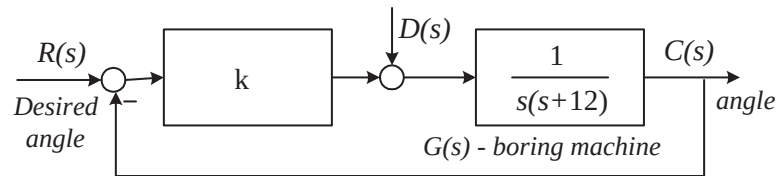


Figure 2.4: Boring machine control system

1. Determine the response $C(s)$ when the input $R(s)$ and the disturbance $D(s)$ are unit step signals.
2. Calculate the steady-state error when
 - a) $d(t) = 0$
 - b) $r(t) = 0$.

3. Determine the values of the gain k for which the overshoot of the response $c(t)$ is zero.
4. Select a gain k such that the overshoot of the closed-loop system is less than 2% and the steady-state is attained in one second, when $r(t)$ and $d(t)$ are unit step signals.

P. 8 A robot arm and camera could be used to pick fruit as shown in Figure 2.5. The camera is used to close the feedback loop to a microcomputer, which controls the arm. The process is:

$$G(s) = \frac{k}{(s + 3)^2}$$

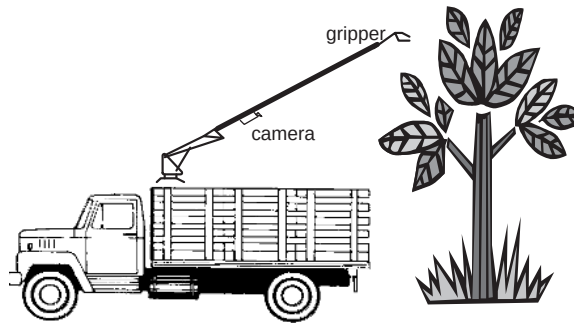


Figure 2.5: Robot fruit picker

1. Calculate the expected steady-state error of the gripper for a step command as function of k .
2. Name a possible disturbance signal for this system.

P. 9 A magnetic disk drive requires a motor to position a read/write head over tracks of data on a spinning disk. The motor and head may be represented by:

$$G(s) = \frac{10}{s(\tau s + 1)}$$

where $\tau=0.001$ second. The controller takes the difference of the actual and desired positions and generates an error. This error is multiplied by an amplifier k .

1. What is the steady-state position error for a step change in the desired input?
2. Calculate the required k in order to yield a steady-state error of 0.1 mm for a ramp input of 10 cm/s.

P. 10 A machine tool is designed to follow a desired path so that

$$r(t) = 2 - t + 0.5t^2.$$

The machine tool control system is shown in Figure 2.6.

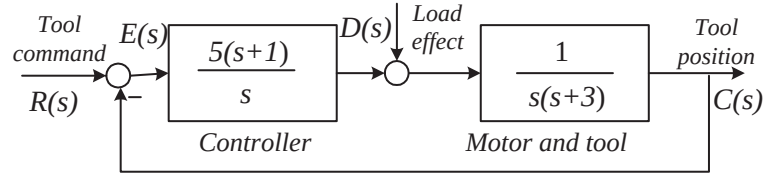


Figure 2.6: Machine tool control system

1. Determine the steady-state error when $r(t)$ is the desired path as given, and $D(s) = 0$.
 2. Plot the error $e(t)$ for the desired path for part (1), for $0 \leq t \leq 10$ seconds.
 3. If the desired input is $r(t) = 0$, find the steady-state error when $D(s) = 1/s$.
 4. Plot the error for part (3) for $0 \leq t \leq 10$ seconds.
- P. 11** The control of the roll angle of an airplane is achieved by using the torque developed by the ailerons, as shown in Figure 2.7.

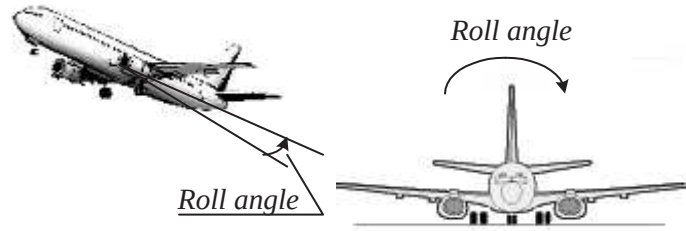


Figure 2.7: Roll angle of an airplane

A linear model of the roll control system for a small experimental aircraft is shown in Figure 2.8 where $q(t)$ is the flow into a hydraulic cylinder and

$$G(s) = \frac{1}{s^2 + 4s + 9}$$

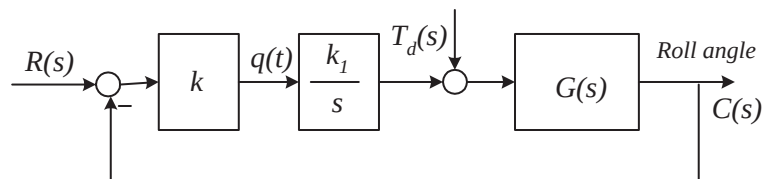


Figure 2.8: Airplane roll control system

The goal is to maintain a small roll angle θ due to disturbances. Select an appropriate gain kk_1 that will reduce the effect of the disturbance while attaining a desirable response to a step disturbance with $r(t) = 0$. To require a desirable transient response, let $kk_1 < 35$.

P. 12 A feedback control system is depicted in Figure 2.9.

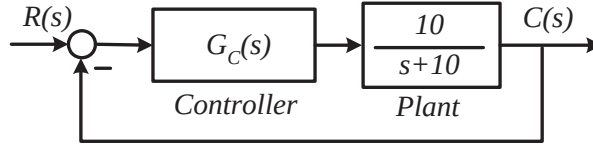


Figure 2.9: Closed-loop control system

Suppose that our design objective is to find a controller of minimal complexity such that the closed-loop system can track a unit step with zero steady-state error.

1. As a first try, consider a simple proportional controller

$$G_C(s) = k$$

where k is a fixed gain. Let $k = 2$. Plot the unit step response and determine the steady-state error from the plot.

2. Consider a more complex controller

$$G_C(s) = k_0 + \frac{k_1}{s}$$

where $k_0 = 2$ and $k_1 = 20$. This controller is known as proportional-integral (PI) controller. Plot the unit step response and determine the steady-state error from the plot.

3. Compare the results from parts (1) and (2) and discuss the trade-off between the controller complexity and the steady-state error performance.

P. 13 When the system shown in Figure 2.10 is subjected to a unit step input, the system responds as shown in Figure 2.11. Determine the values k and T from the response curve. Check the results plotting the unit step response of the system obtained.

P. 14 Determine the values of k_1 and k_2 of the closed-loop system shown in Figure 2.12 so that the maximum overshoot in unit step response is 25% and the peak time is 3 seconds. Assume $J = 1 \text{ kg} \cdot \text{m}^2$.

P. 15 Consider the system shown in Figure 2.13 a).

1. Determine the steady-state error to a unit ramp input.

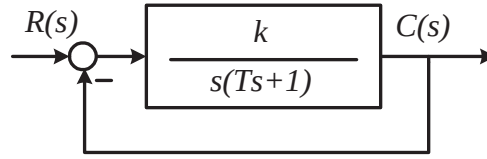


Figure 2.10: Closed-loop control system

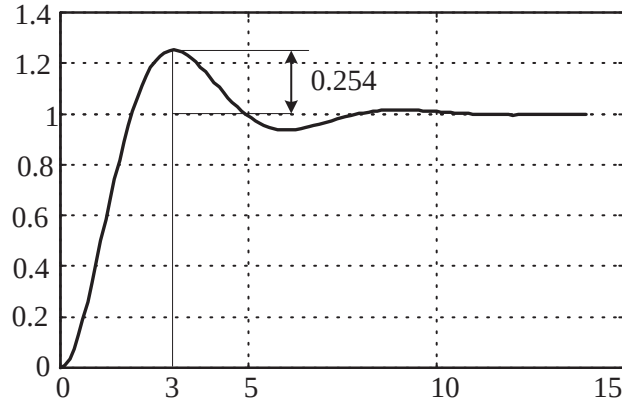


Figure 2.11: Unit step response curve

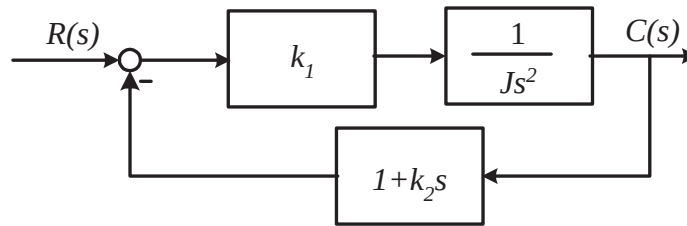


Figure 2.12: Closed-loop control system

2. Show that the steady-state error for following a ramp input may be eliminated if the input is introduced to the system through a proportional plus derivative filter, as shown in Figure 2.13 b) and the value of k is properly set. Note that the error $e(t)$ is given by $r(t) - c(t)$.

P. 16 Consider a unity-feedback control system whose open-loop transfer function is:

$$G(s) = \frac{k}{s(Js + B)}$$

Discuss the effects that varying the values of k and B has on the steady-state error in unit-ramp response. Plot the unit ramp response curves for small value, medium value and large value of k .

P. 17 A low inertia plotter is shown in Figure 2.14. This system may be represented by the block diagram shown in Figure 2.15.

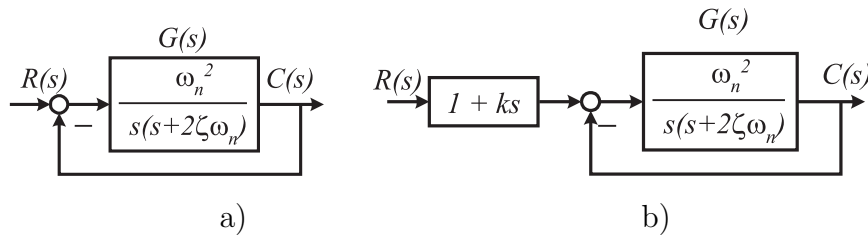


Figure 2.13: Closed-loop control system

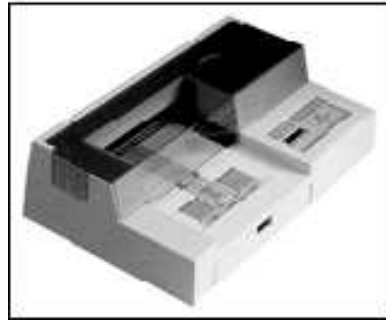


Figure 2.14: Plotter

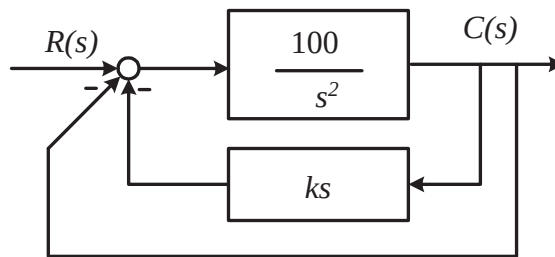


Figure 2.15: Block diagram of plotter

1. Calculate the steady-state error for a ramp input.
2. Select a value for k that will result in zero overshoot to a step input but as rapid response as is attainable. Use Matlab in simulations.

P. 18 A control system for positioning the head of a floppy disk has the closed-loop transfer function

$$T(s) = \frac{0.313(s + 0.8)}{(s + 0.25)(s^2 + 0.3s + 1)}$$

Plot the poles and zeros of this system and discuss the dominance of the complex poles. What overshoot for a step input do you expect?

P. 19 A second-order control system has the closed-loop transfer function $T(s) = C(s)/R(s)$. The system specifications for a step input follow:

1. Percent overshoot $\leq 5\%$.
2. Settling time < 4 seconds.
3. Peak time $T_p < 1$ second.

Show the permissible area for the poles of $T(s)$ in order to achieve the desired response.

P. 20 For the feedback system shown in Figure 2.16:

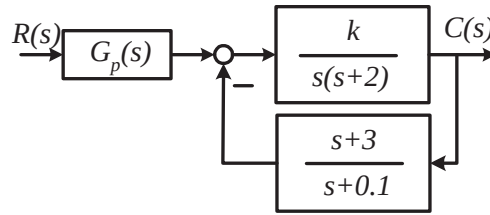


Figure 2.16: Closed loop system

1. Determine the steady-state error for a unit step when $k = 0.4$ and $G_p(s) = 1$.
2. Select an appropriate value for $G_p(s)$ so that the steady-state error is zero for a step input.

P. 21 The roll control autopilot of a jet fighter is shown in Figure 2.17. The goal is to

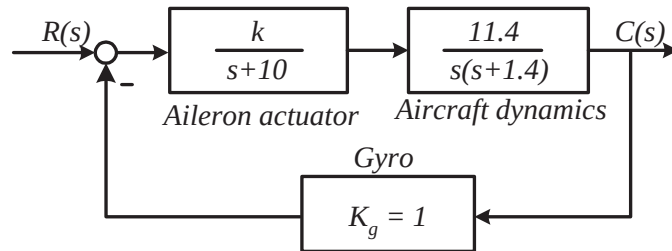


Figure 2.17: Closed loop system

select a suitable k such that the response to a unit step command $r(t) = 1, t \geq 0$, will provide a response $c(t)$ with a fast response and an overshoot of less than 20%.

1. Determine the closed-loop transfer function.
2. Determine the roots of the characteristic equation for $k = 0.7, 3$ and 6 .
3. Using the concept of dominant roots, find the expected overshoot and peak time for the approximate second-order system.
4. Plot the actual response and compare with the approximate results of part (3).

5. Select the gain k so that the percent overshoot is equal to 16%. what is the resulting peak time?

P. 22 The model for a position control system using a d.c. motor is shown in Figure 2.18. The goal is to select k_1 and k_2 so that the peak time is 0.8 second and the percent

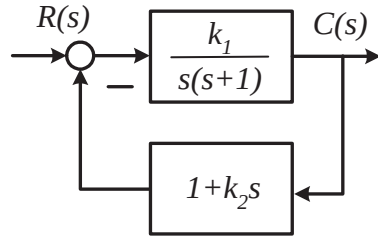


Figure 2.18: Position control system

overshoot (PO) for a step input is negligible ($0.5\% < PO < 2\%$).

P. 23 Consider the control system shown in Figure 2.19.

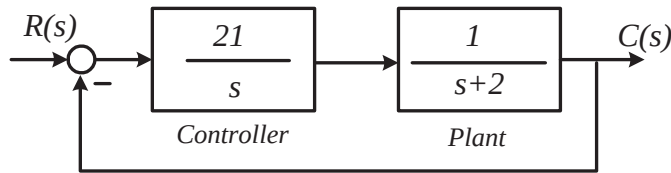


Figure 2.19: Control system

1. Show analytically that the expected percent overshoot of the closed-loop response to a step input is about 50%.
2. Plot the unit step response of the closed-loop system and estimate the percent overshoot from the plot. Compare the result with part (1).

P. 24 Consider an electric furnace whose temperature is controlled by the voltage V_r . In order to determine the furnace behaviour, an experimental trial is done. The experiment consists on measuring the temperature inside the furnace T_i for a constant feed voltage $V_r = 1V$. The result of measurement is shown in Figure 2.20.

1. Identify the transfer function $T_i(s)/V_r(s)$.
2. The furnace has to be heated in a progressive manner until the temperature reaches $100^\circ C$. The feed voltage is applied as a unit ramp. determine the time necessary for the temperature to reach $100^\circ C$.

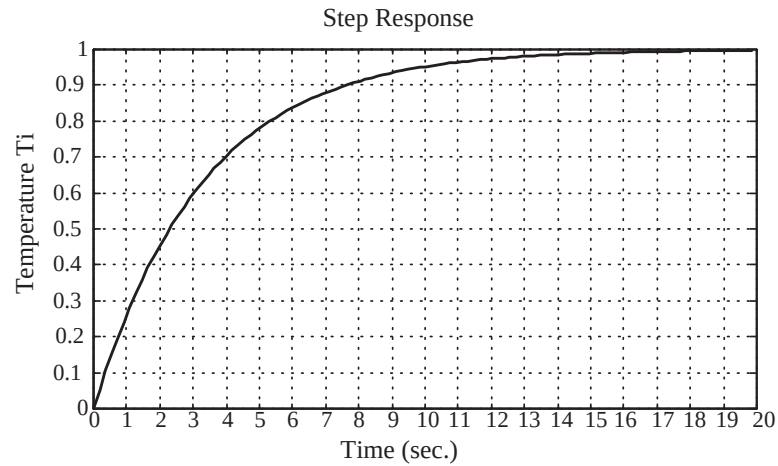


Figure 2.20: Step response of the furnace

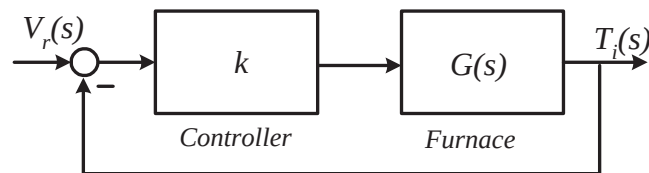


Figure 2.21: Furnace control system

3. To improve the furnace behaviour a control system shown in Figure 2.21 is used. The gain k can be adjusted between 0 and 100. Discuss the system behaviour in this case. Is it better to use a big value for k or a small one?

P. 25 Consider the closed-loop control system with the block diagram shown in Figure 2.22. Determine the unit step response for $k = 0.02$, $k = 0.125$ and $k = 2.5$ and compare

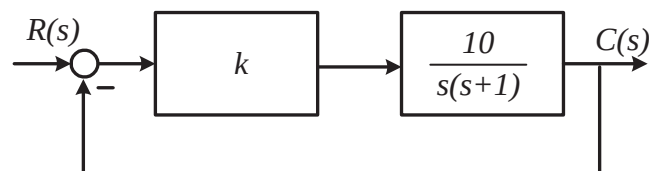


Figure 2.22: Control system

the peak times, settling times and percent overshoots.

2.2 The stability of linear systems

P. 1 Utilizing the Routh-Hurwitz criterion determine the stability of the following polynomials:

1. $s^2 + 4s + 1$
2. $s^3 + 4s^2 + 6s + 6$
3. $s^3 + 3s^2 - 6s + 10$
4. $s^4 + s^3 + 2s^2 + 5s + 8$
5. $s^4 + s^3 + 3s^2 + 2s + k$
6. $s^5 + s^4 + 2s^3 + s^2 + s + k$

P. 2 Consider the closed-loop feedback system shown in Figure 2.23

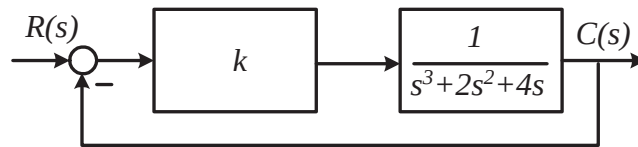


Figure 2.23: Closed-loop control system

1. Using the Routh-Hurwitz criterion find the range of k for stability.
2. Plot the roots of the characteristic equation for $0 \leq k \leq 20$ and show that the system is stable for k determined at (1).

P. 3 A system has a characteristic equation

$$s^3 + 3ks^2 + (2 + k)s + 4 = 0$$

Determine the range of k for a stable system.

P. 4 A feedback control system is shown in Figure 2.24.

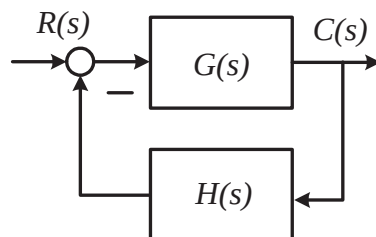


Figure 2.24: Closed-loop control system

The process transfer function is

$$G(s) = \frac{k(s+40)}{s(s+10)}$$

and the feedback transfer function is

$$H(s) = \frac{1}{s+20}$$

1. Determine the limiting value of gain k for a stable system.
 2. For the gain that results in marginal stability determine the magnitude of the imaginary roots.
- P. 5** A control system is shown in Figure 2.25. It is desired that the system is stable and

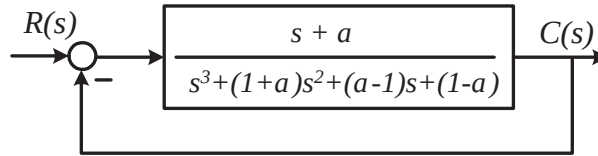


Figure 2.25: Closed-loop control system

the steady-state error for a unit step input be less than or equal to 0.05.

1. Determine the range of a that satisfies the error requirement.
 2. Determine the range of a that satisfies the stability requirement.
 3. Select an a that meets both requirements.
- P. 6** A closed-loop control system is shown in Figure 2.26.

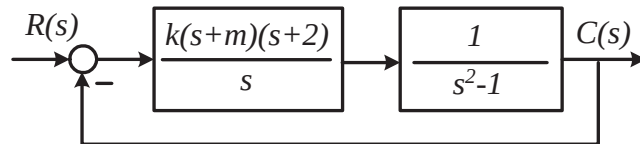


Figure 2.26: Closed-loop control system

1. Determine the range of gain k and parameter m so that the system is stable, and plot the region of stability.
2. Select the gain and parameter m values so that the steady-state error to a ramp input is less than or equal to 10% of the input magnitude.
3. Determine the percent overshoot for a step input for the design selected in part (2).

P. 7 Consider the feedback control system shown in Figure 2.27 where

$$G(s) = \frac{k}{s(s+1)(s+2)}$$

For $0 \leq k \leq 10$ compute and plot the closed-loop transfer function poles. Determine

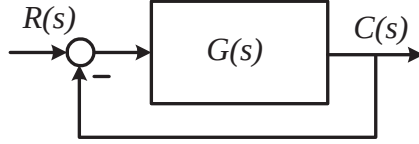


Figure 2.27: Closed-loop control system

the maximum range of k for stability. Compute the roots of the characteristic equation where k is the minimum value allowed for stability.

P. 8 Consider the closed-loop control system shown in Figure 2.28.

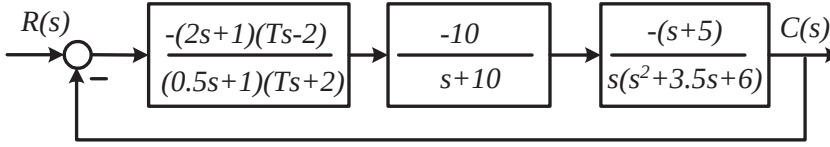


Figure 2.28: Closed-loop control system

The variable T represents a time delay. Compute the closed-loop system poles for a slow system ($T = 0.5$) and a fast one ($T = 0.25$). Comment on the results. What is the maximum time delay allowable for stability?

P. 9 Consider the system shown in Figure 2.29.

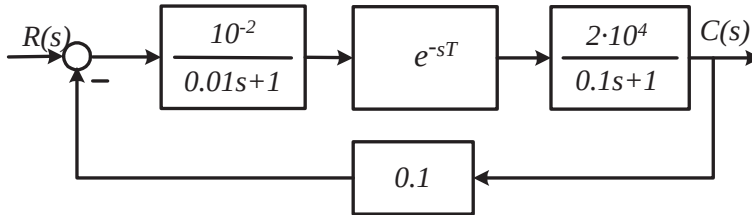


Figure 2.29: Closed-loop control system

1. Determine the closed-loop transfer function $C(s)/R(s)$.
2. Approximate the exponential function by the Pade approximation:

$$e^{-sT} = \frac{1 - sT/2}{1 + sT/2}$$

and obtain the maximum value of time delay T for a stable system.

P. 10 Robot-Controlled Motorcycle. Consider the robot controlled motorcycle shown in Figure 2.30. The motorcycle will move in a straight line at constant forward speed V .

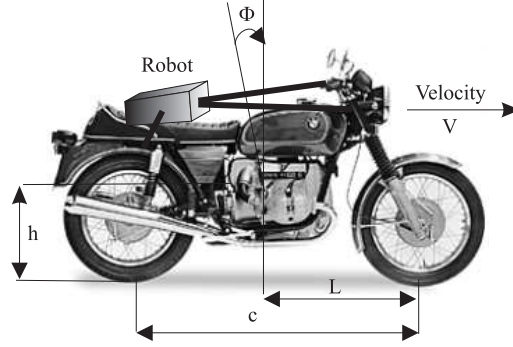


Figure 2.30: Robot controlled motorcycle

Let Φ denote the angle between the plane of symmetry of the motorcycle and the vertical. The desired angle Φ_d is equal to zero.

$$\Phi_d(s) = 0$$

The goal is to maintain the motorcycle in the vertical position, and maintain the prescribed position in the presence of disturbances. The variable to be controlled is the motorcycle position from vertical (Φ) such that the closed-loop system is stable.

The motorcycle simplified model is given by:

$$G(s) = \frac{1}{s^2 - a_1^2}$$

where $a_1 = g/h$, $g = 9.806 \text{ m/sec}^2$, and $h = 1.09 \text{ m}$ is the height of the motorcycle center of gravity above the ground.

The controller is given by:

$$G_C(s) = \frac{a_2 + a_3 s}{Ts + 1}$$

where

$$a_2 = V^2/(hc),$$

$$a_3 = VL/(hc).$$

The forward speed of the motorcycle, denoted by V is equal to 2 m/sec. The distance between wheel centers is $c = 1.36 \text{ m}$. The horizontal distance between the front wheel axle and the motorcycle center of gravity is $L = 1 \text{ m}$. The time constant of the controller is denoted by $T = 0.2 \text{ sec}$.

Control is accomplished by turning the handlebar. The front wheel rotation about the vertical is not evident in the transfer function. Also the transfer function assume a constant forward speed V , which means we must have another control system at work regulating the forward speed.

Assembling the components of the feedback system gives us the system configuration shown in Figure 2.31.

The parameters we want to adjust are in the feedback loop.

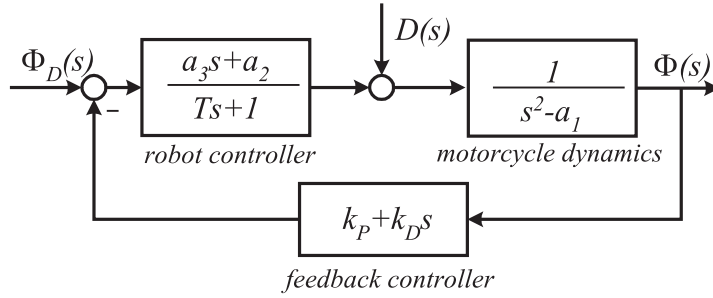


Figure 2.31: Block diagram of robot motorcycle control system

1. Use the Routh-Hurwitz technique to analyze the closed-loop system stability. Determine the values of k_P and k_D which lead to the closed-loop stability. Plot the stability region as a function of k_P and k_D . Select two values for k_P and k_D from the stability region and show that the closed-loop poles are stable.
2. Analyze the stability of the open-loop system to unit step disturbances. (The transfer function for the disturbance $D(s)$ to the output without feedback is $G(s)$).
3. For the parameters selected in part (1) show that the closed-loop control system of the motorcycle is stable. Plot the system response for unit step disturbance.

Chapter 3

Root Locus

3.1 Root locus analysis

P. 1 Sketch the root loci of the following pole-zero maps shown in Figure 3.1. Show the asymptotes.

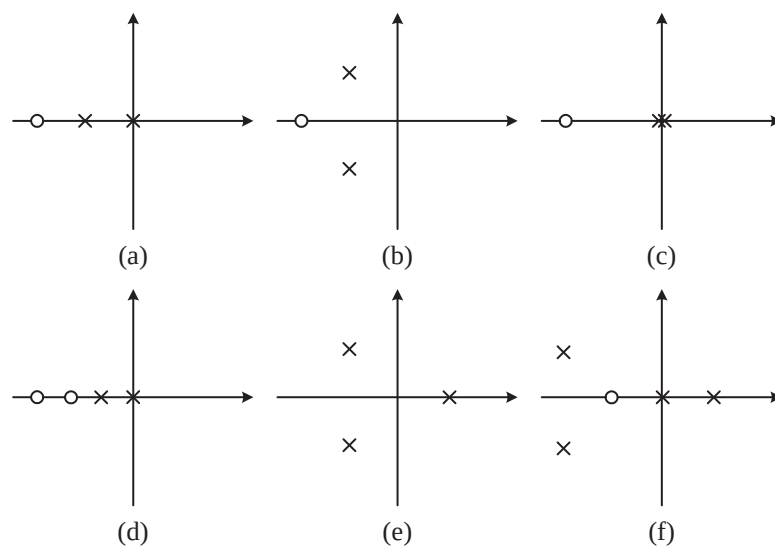


Figure 3.1: Pole-zero maps

P. 2 Draw the root locus for the following open-loop transfer functions:

$$GH(s) = \frac{k}{s(s+1)^2}$$

$$GH(s) = \frac{k}{(s^2 + 2s + 2)(s+2)}$$

$$GH(s) = \frac{k(s+1)}{s(s+2)(s+3)}$$

$$GH(s) = \frac{k(s+2)(s+3)}{(s+1)(s^2 + 2s + 2)}$$

P. 3 Consider the closed-loop control system with the characteristic equation:

$$1 + \frac{ks(s+4)}{s^2 + 2s + 2} = 0$$

1. Sketch the root locus.
2. Find the gain where the roots are both equal.
3. Find these two roots.
4. Find the settling time of the system when the roots are equal.

P. 4 Sketch the root locus for the following characteristic equation:

$$1 + \frac{k(s+1)}{s(s+4)(s+5)} = 0$$

Determine the maximum value of k for which all the closed-loop poles are real.

P. 5 A control system as shown in Figure 3.2 has a plant:

$$G(s) = \frac{1}{s(s-1)}$$

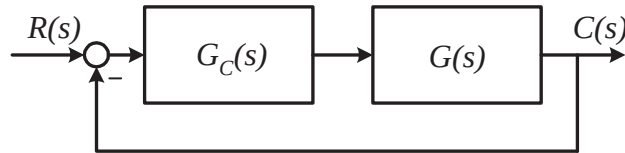


Figure 3.2: Closed-loop control system

1. When $G_C(s) = k$, show that the system is always unstable by sketching the root locus.
2. When

$$G_C(s) = \frac{k(s+2)}{s+20}$$

sketch the root locus and determine the range of k for which the system is stable. Determine the value of k and the complex roots when two complex roots lie on the $j\omega$ axis.

P. 6 A unity feedback system has a plant

$$G(s) = \frac{k(s+1)}{s(s-1)(s+4)}$$

1. Determine the range of k for stability.
2. Draw the root locus.
3. Determine the maximum damping factor ζ of the stable complex roots.

P. 7 A unity negative feedback system has the open-loop transfer function:

$$k \cdot G(s) = k \cdot \frac{s^2 - 2s + 2}{s(s^2 + 3s + 2)}$$

Plot the root locus and show that the maximum value of k for a stable system is $k = 0.79$.

P. 8 Consider the feedback control system in Figure 3.3.

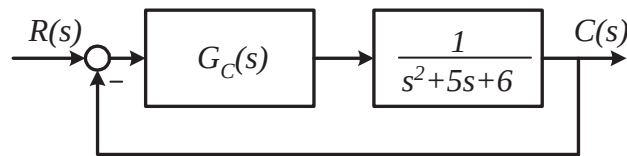


Figure 3.3: Closed-loop control system

We have three potential controllers for our system:

- $G_C(s) = k$ - proportional (P) controller.
- $G_C(s) = k/s$ - integral (I) controller.
- $G_C(s) = k(1 + 1/s)$ - proportional-integral (PI) controller.

The design specifications for a unit step input are:

- settling time $t_s \leq 10$ seconds, and
- percent overshoot $M_p \leq 10\%$.

1. For the P-controller draw the root locus for $0 \leq k \leq \infty$ and determine k such that the design specifications are satisfied.
2. Repeat part (1) for the I-controller.
3. Repeat part (1) for the PI-controller.
4. Co-plot the unit step responses for the closed-loop systems with each controller designed in parts (1) - (3).
5. Compare and contrast the controllers obtained in parts (1) - (3), concentrating the discussion on the steady-state errors and transient performance.

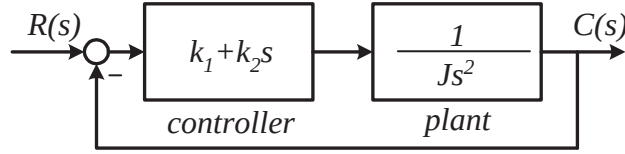


Figure 3.4: Closed-loop control system

P. 9 Consider the control system shown in Figure 3.4.

The controller is known as a proportional-derivative (PD) controller. Suppose that we require the ratio of $k_1/k_2 = 5$. Using the root locus method find the values of k_2/J and k_1/J such that the settling time t_s is less or equal than 4 seconds and the percent overshoot M_p is less or equal than 10% for a step input.

P. 10 A unity feedback system has an open-loop transfer function:

$$G(s) = \frac{k(s^2 + 3s + 3.25)}{s^2(s + 1)(s + 10)(s + 20)}$$

Draw the root locus for $k \geq 0$ and select a value for k that will provide an acceptable step response (minimum overshoot, short settling time).

P. 11 A unity feedback control system is shown in Figure 3.5. Draw the root locus

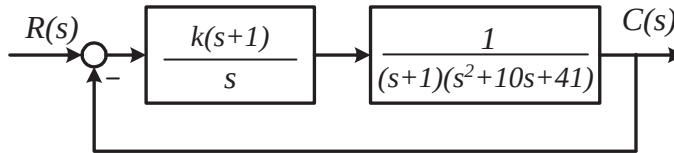


Figure 3.5: Closed-loop control system

and predict the step response of the system for a selected gain k . Determine the actual response of the system and compare it to the predicted response.

P. 12 A feedback system is shown in Figure 3.6. Draw the root locus as k varies when

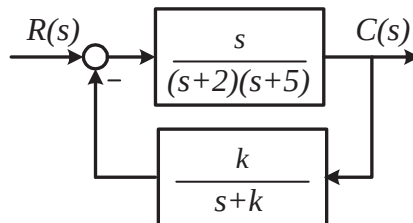


Figure 3.6: Closed-loop control system

$k \geq 0$. Determine a value for k that will provide a step response with an overshoot less than 5% and a settling time less than 2.5 seconds.

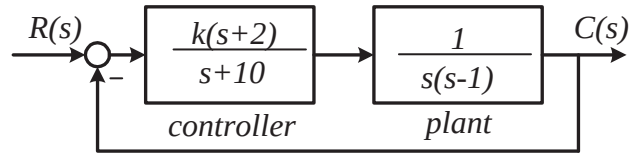


Figure 3.7: Closed-loop control system

P. 13 A model of a system that is open-loop unstable is shown in Figure 3.7. Using a root locus, select k for the maximum achievable damping ratio ζ , of the complex roots. Predict the step response of the system and compare it with the actual step response.

P. 14 For the system shown in Figure 3.8 find the value of k ($k \geq 0$), for which the percent overshoot of the closed-loop system is less than 10%.

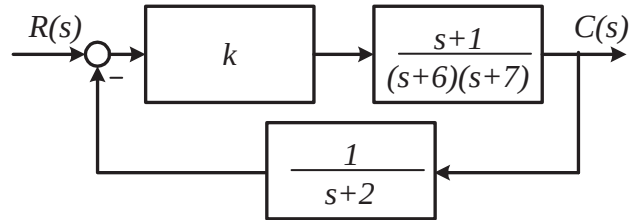


Figure 3.8: Closed-loop control system

P. 15 Sketch the root locus for the system shown in Figure 3.9. Determine the value of

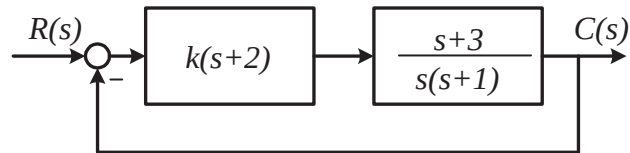


Figure 3.9: Closed-loop control system

k for which the system is overdamped and underdamped.

P. 16 Plot the root locus for a system with the open-loop transfer function

$$G(s)H(s) = \frac{k(s+1)}{s(s-1)(s^2+4s+16)}$$

Find the range of gain k for stability.

P. 17 Consider the system shown in Figure 3.10. Determine the value of a such that the damping ratio ζ of the dominant closed-loop poles is 0.5.

P. 18 Consider the system shown in Figure 3.11. Sketch the root locus as a varies from 0 to ∞ . Determine the value of a such that the damping ratio value of dominant closed-loop poles is 0.5.

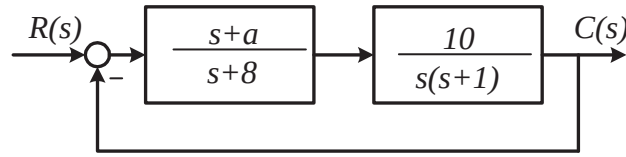


Figure 3.10: Closed-loop control system

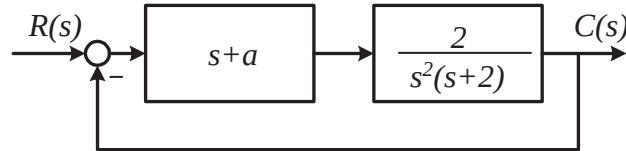


Figure 3.11: Closed-loop control system

P. 19 A unity feedback system has a forward transfer function:

$$G(s) = \frac{k(s+10)}{s(s^2+6s+8)}$$

Sketch the root locus diagram for the positive values of k . Use this sketch to determine:

1. The maximum value of k for which the system has no complex roots
2. The value of k for which the complex poles have a damping ratio of 0.6.

Use Matlab to produce an accurate root locus plot and repeat the above calculations. How do these answers compare with those obtained from your sketched diagram?

Chapter 4

Analysis in Frequency Domain

4.1 Bode diagrams

P. 1 Sketch the Bode diagram representation of the frequency response for the following transfer functions:

$$GH(s) = \frac{1}{(1 + 0.5s)(1 + 2s)}$$

$$GH(s) = \frac{1 + 0.5s}{s^2}$$

$$GH(s) = \frac{s + 4}{s^2 + 5s + 25}$$

$$GH(s) = \frac{30(s + 8)}{s(s + 2)(s + 4)}$$

P. 2 A bandpass amplifier has the transfer function:

$$G(s) = \frac{10^9 s}{(s + 1000)(s + 10^7)}$$

1. Sketch the Bode diagram for $G(j\omega)$.
2. Find the midband gain (in dB).
3. Find the high and low frequency -3 dB points.

P. 3 A spring-mass-damper system is shown in Figure 4.1. The linear differential equation which describes the system is:

$$M \frac{d^2 y}{dt^2} + f \frac{dy}{dt} + ky = r(t)$$

where M is the mass, f - the friction coefficient, k - the spring constant, $y(t)$ - the displacement of the mass, and $r(t)$ - the input force. The Bode diagram obtained by experimental means using a sinusoidal forcing function $r(t)$, is shown in Figure 4.2. Determine the numerical values of M , f and k .

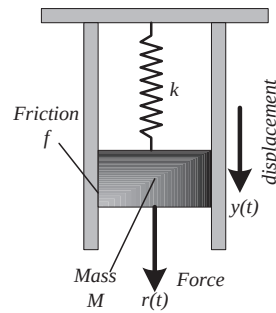


Figure 4.1: Spring-mass-damper system

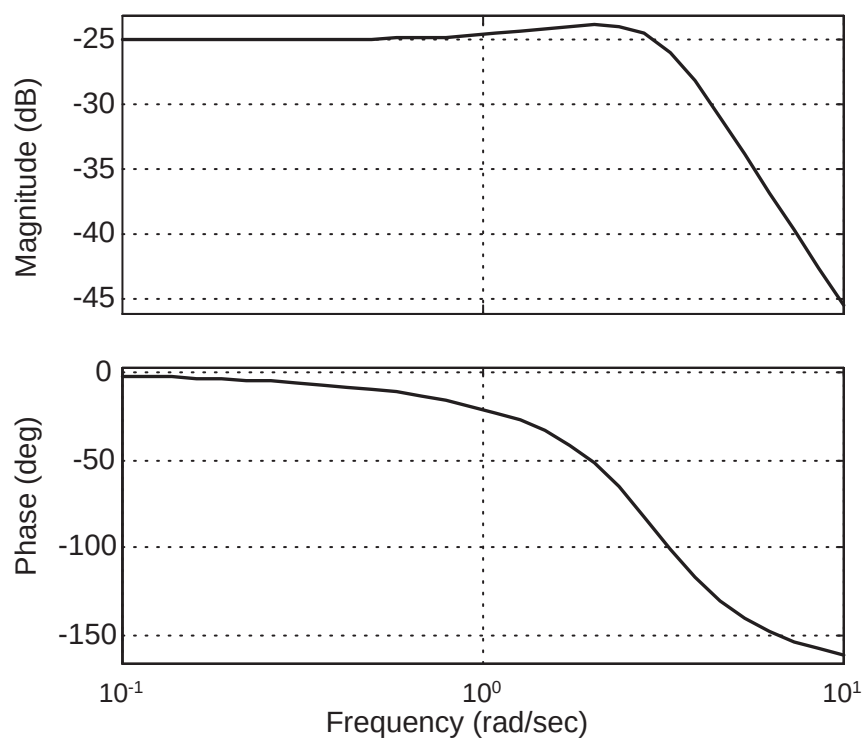


Figure 4.2: Spring-mass-damper system

P. 4 Consider a system with the transfer function:

$$G(s) = \frac{25}{s^2 + s + 25}$$

Obtain the Bode plot and verify that the resonant frequency is 5 rad/sec and the resonant peak magnitude $M_{p\omega}$ is 14 dB. Determine the bandwidth ω_B .

P. 5 A unity negative feedback system has the open-loop transfer function:

$$G(s) = \frac{100}{s(s+6)}$$

Determine the resonant peak $M_{p\omega}$, the resonant frequency ω_r and the bandwidth ω_B of the system from the closed-loop Bode plot.

P. 6 A system has a transfer function

$$G(s) = \frac{4(1 + 1/3s)}{s(1 + 2s)(1 + 1/7s + 1/49s^2)}$$

Plot the Bode diagram and find the phase margin and the gain margin.

P. 7 Consider a system with the open-loop transfer function

$$G(s) = \frac{20(s+1)}{s(s^2 + 2s + 10)(s+5)}$$

Draw a Bode diagram and determine the phase margin and the gain margin.

P. 8 A system has an open-loop transfer function

$$G(s) = \frac{k(s+100)}{s(s+10)(s+40)}$$

Use Bode diagram to show that for $k = 500$ the system is unstable. Determine the resonant peak and the phase margin for $k = 50$.

P. 9 A unity feedback system has a plant

$$G(s) = \frac{k(s+3)}{s(s+1)(s+5)}$$

Plot the Bode diagram and determine the gain k required to obtain a phase margin of 40° .

P. 10 A closed-loop system has the block diagram shown in Figure 4.3. Plot the Bode diagram of the open-loop transfer function. Determine the gain and phase margin of this system.

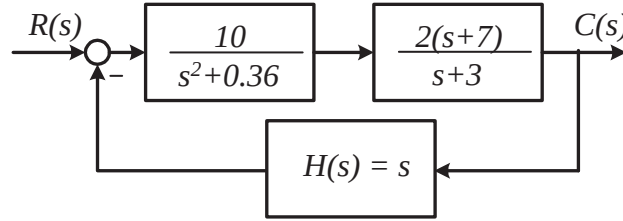


Figure 4.3: Closed-loop system

P. 11 A closed-loop system has a loop transfer function

$$GH(s) = \frac{k}{s(s+1)(s+10)}$$

Determine the gain k so that the phase margin is 60° . For the gain selected, determine the gain margin of the system.

P. 12 A closed-loop with unity feedback system has a loop transfer function

$$GH(s) = \frac{k(s+4)}{s^2}$$

Determine the gain k so that the phase margin of 60° . For the gain selected, determine the gain margin.

4.2 Nyquist diagrams

P. 1 Plot the Nyquist diagram for the following transfer functions:

$$\begin{aligned} G(s) &= \frac{1}{s^2 + s + 1} \\ G(s) &= \frac{10}{s^2 + s + 1} \\ G(s) &= \frac{20(s^2 + s + 0.5)}{s(s+1)(s+10)} \end{aligned}$$

P. 2 Show that the Nyquist plot of

$$G(s) = \frac{1}{s+a}$$

is a semicircle of radius $1/(2a)$ and centre $(1/(2a), 0)$.

P. 3 The polar plot of the open-loop frequency response of a unity-feedback control system is shown in Figure 4.4. Assuming that the Nyquist path in the s -plane encloses the entire right-half s -plane, draw a complete Nyquist plot in the G -plane. Then answer the following questions:

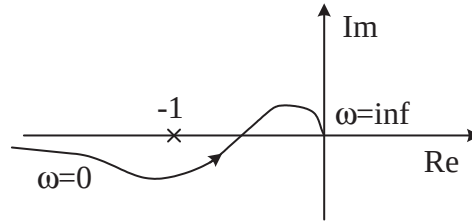


Figure 4.4: Nyquist plot of the open-loop transfer function

1. If the open-loop transfer function has no poles in the right-half s-plane, is the closed-loop stable?
 2. If the open-loop transfer function has one pole and no zeros in the right-half s-plane, is the closed-loop system stable?
 3. If the open-loop transfer function has one zero and no poles in the right-half s-plane, is the closed-loop system stable?
- P. 4** Plot the polar plots of the following open-loop transfer functions and determine whether the system is stable by utilizing the Nyquist criterion:

$$GH(s) = \frac{k}{s(s^2 + s + 4)}$$

$$GH(s) = \frac{k(s + 2)}{s(s + 4)}$$

If the system is stable, find the maximum value for k by determining the point where the polar plot crosses the real axis.

- P. 5** A negative unity feedback system has the open-loop transfer function

$$GH(s) = \frac{k}{s(s + 1)(2s + 1)}$$

Find the critical value of gain k for stability by utilizing the Nyquist criterion.

- P. 6** Consider a negative unity feedback system with the loop transfer function

$$G(s) = \frac{k}{s - 1}$$

Determine the critical value of k for stability by use of the Nyquist stability criterion.

- P. 7** Draw a Nyquist locus for the unity feedback control system with the open-loop transfer function:

$$G(s) = \frac{k(1 - s)}{s + 1}$$

P. 8 A system with the open-loop transfer function

$$GH(s) = \frac{k}{s^2(Ts + 1)}$$

is inherently unstable. This system can be stabilized by adding derivative control. Draw the polar plots for the open-loop transfer function with and without derivative control for $k = 0.2$, $T = 0.1$.

P. 9 Consider the unity feedback control system whose open-loop transfer function is

$$G(s) = \frac{as + 1}{s^2}$$

Determine the value of a so that the phase margin is 45° .

Chapter 5

Control System Design

5.1 Problems

P. 1 Consider the control system that models a jet fighter, shown in Figure 5.1.

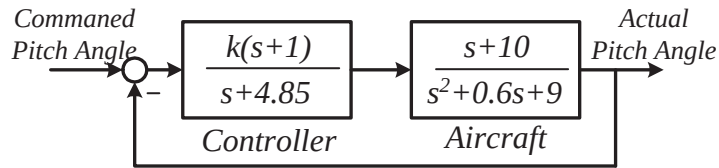


Figure 5.1: Control system

1. What is the range of k for stability?
 2. In general, the pilot would like the actual pitch angle to be very close to his commanded pitch angle. How should one choose k to make this happen, and why?
- P. 2** Consider a plant with the transfer function:

$$G(s) = \frac{1}{s(s^2 + s + 4)}$$

Implement a unity-feedback control-loop for this plant.

1. Draw a block diagram of the control system showing your controller, the plant, the setpoint $r(t)$, the measured output $c(t)$, the controller output $u(t)$ and the error $e(t) = r(t) - c(t)$
2. Build the controller. Consider the controller output given by the following equations:
 - (a) P controller: $u(t) = 50e(t)$
 - (b) PD controller: $u(t) = 5e(t) + 20 \frac{de(t)}{dt}$
 - (c) PI controller: $u(t) = 6e(t) + 20 \int_0^t e(\tau) d\tau$

Find the controller transfer function (from e to u), the transfer function of the closed-loop system (from r to c), the stability of the closed-loop system and the steady-state error to a step input. Which of the three controllers would you use and why?

3. If the reference input is a unit ramp not a step can you use the same design? If not, what modifications would you make?

P. 3 Consider a plant having the transfer function:

$$G(s) = \frac{1}{s^2(s+2)}$$

Plot the root locus for the following controllers. The closed-loop settling time should be less than 10 seconds. Sketch in the acceptable regions of the s-plane for the closed-loop poles. If it is possible to satisfy these criteria for some gain k , choose a gain by placing a 'x' on the root locus where you would place the closed-loop poles of the system.

1. Proportional control: $G_c(s) = k$. Can the criteria be satisfied for proportional control?
2. PD control: $G_c(s) = k(s+0.5)$. Can the criteria be satisfied for PD control?

P. 4 Consider a simple spring, mass and damper problem as shown in Figure 5.2.

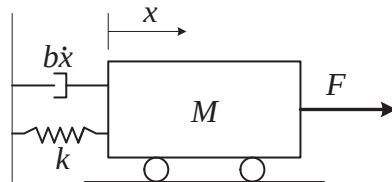


Figure 5.2: Spring-mass-damper system

The modeling equation of this system is:

$$M\ddot{x}(t) + b\dot{x}(t) + kx(t) = F(t)$$

where: $M = 1$, $b = 10$, $k = 20$, $F(t) = 1$.

1. Obtain the transfer function $G(s)$ of this system.
2. Plot the open-loop unit step response and determine the steady-state error.
3. Consider a unity-feedback control system as shown in Figure 5.3 with an ideal PID-controller and the plant $G(s)$. The transfer function of the PID-controller is:

$$G_{PID} = K_P + \frac{K_I}{s} + K_D s$$

Show you how each of K_P , K_I and K_D contributes to obtain:

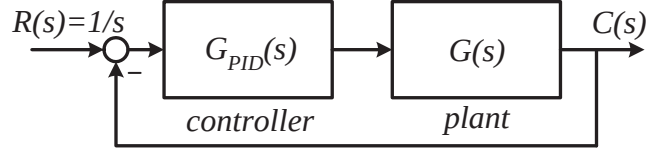


Figure 5.3: Closed-loop control system

- Fast rise time
- Minimum overshoot
- Minimum or zero steady-state error

Consider first a proportional (P) controller ($K_I = K_D = 0$) and determine the influence of K_P on the rise time, overshoot and steady-state error. Then look at the influence of the derivative term K_D term by taking a proportional-derivative (PD) controller ($K_I = 0$), and finally consider a proportional-integral (PI) controller ($K_D = 0$) and determine the effect of K_I .

P. 5 A negative feedback control system has a transfer function:

$$G(s) = \frac{k}{s+3}$$

We select a compensator

$$G_c(s) = \frac{s+a}{s}$$

in order to achieve zero steady-state error for a step input. Select a and k so that the overshoot to a step input is approximately 5% and the settling time is approximately 1 second.

P. 6 Consider the system shown in Figure 5.4. Design a controller $G_C(s)$, as simple as

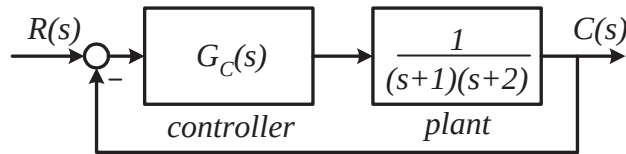


Figure 5.4: Control system

possible, such that for a unit step input, the closed-loop system has:

- the percent overshoot $M_p \leq 5\%$,
- the settling time $t_s \approx 2.1s$,
- the steady-state error $\varepsilon_{st} \leq 50\%$.

P. 7 For the system shown in Figure 5.4 design a controller $G_C(s)$ such that:

- the percent overshoot $M_p \leq 5\%$,
- the settling time $t_s \approx 2.1s$,
- the steady-state error $\varepsilon_{st} \leq 10\%$.

P. 8 For the system shown in Figure 5.4 design the simplest controller $G_C(s)$ for which:

- the percent overshoot $M_p \leq 5\%$,
- the settling time $t_s \approx 1s$,
- the steady-state error $\varepsilon_{st} \leq 25\%$.

P. 9 Given the following unity feedback system with:

$$G(s) = \frac{k}{(s+1)(s+4)}$$

design a lead compensator that will yield a peak time of 1.047 seconds and a damping ratio of 0.8. Calculate the gain at that value.

P. 10 The forward transfer function $G(s)$ for the system in Figure 5.5 is

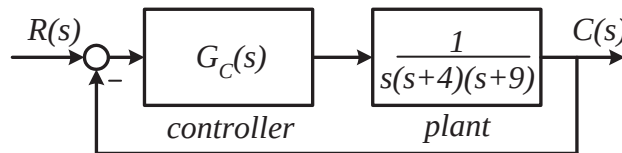


Figure 5.5: Control system

$$G(s) = \frac{1}{s(s+4)(s+9)}$$

- Design a compensator $G_c(s)$ of your choice so that the dominant poles of the closed-loop system are located at $-3 \pm 4j$.
- Predict the maximum overshoot and the settling time of the step response of the compensated system based on the second-order approximation. What is the steady-state error of the compensated system to a unit ramp input.

P. 11 Consider the system shown in Figure 5.6. It is desired to design a PID controller $G_c(s)$ such that the dominant closed-loop poles are located at $s = -1 \pm \sqrt{3}$. For the PID controller, choose $a = 0.2$ and then determine the values of k and b . Sketch the root locus of the designed system.

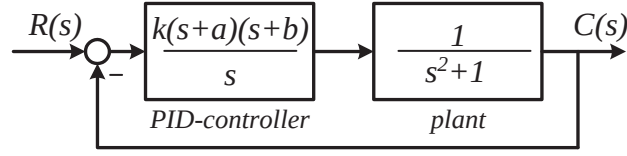


Figure 5.6: PID-controlled system

P. 12 Consider a unity feedback system whose feedforward transfer function is:

$$G(s) = \frac{1}{s(s+2)(s+8)}$$

Design a lead compensator

$$G_c(s) = \frac{k(s+z)}{s+p}$$

so that the dominant closed-loop poles are located at $s = -2 \pm 2\sqrt{3}j$. Now add a lag compensator

$$G_{clag} = \frac{s+zl}{s+pl}$$

such that the static velocity error constant of the compensated system is $K_{vcomp} = 80$.

P. 13 Consider the control system shown in Figure 5.7. Design a lead compensator $G_c(s)$

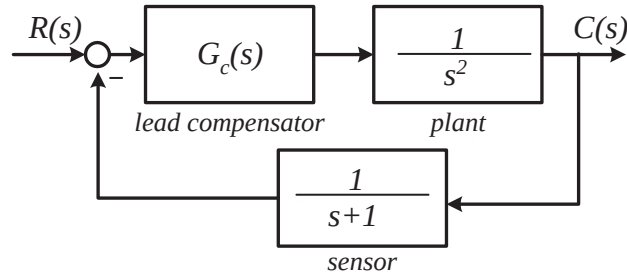


Figure 5.7: Control system

such that the damping ratio ζ and the undamped natural frequency ω_n of the dominant closed-loop poles are 0.5 and 0.2 rad/sec, respectively.

P. 14 Consider the control system shown in Figure 5.8. The plant transfer function is

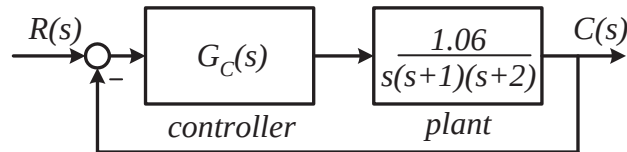


Figure 5.8: Control system

$$G(s) = \frac{1.06}{s(s+1)(s+2)}$$

Determine the dominant closed-loop poles of the uncompensated closed-loop system (when $G_c(s) = 1$). It is desired to increase the static velocity error constant to about 5sec^{-1} without appreciably changing the location of the dominant closed-loop poles. Design a lag compensator $G_c(s)$ to meet this specification.

P. 15 For a unity-feedback closed-loop system with the forward transfer function:

$$G(s) = \frac{k}{(s+1)(s+6)(s+60)}$$

1. Calculate a PI controller that eliminates the steady-state error. Show the results on a step response in Matlab.
2. Calculate a lag controller that gives a 3 fold improvement in steady-state error.

P. 16 Consider unity-feedback closed-loop system with the forward transfer function:

$$G(s) = \frac{20}{s(s+4)(s+40)}$$

1. What is the type number for this system? What is the steady-state error for a unit step input? What is the steady-state error for a unit ramp input?
2. Sketch the uncompensated root locus for this system. How much gain can we put into the forward loop before the system becomes unstable?
3. Design a lead compensator to improve the system response so that the system meet the following specifications:
 - Settling time $t_s = 1$ second
 - Damping ratio of the dominant poles $\zeta=0.707$
 - Steady-state error for a step input, $e_{ss}|_{step}=0$
4. Design a lag compensator to put in series with the lead compensator designed before so that the system meets an additional error requirement:
 - Steady-state error for a ramp input $e_{ss}|_{ramp}=1/20$
5. Sketch both the lead compensated and the lag compensated root loci (use Matlab to help).
6. Find the closed-loop poles of the lead compensated and lag compensated design (i.e. find the roots of $1 + G(s)G_{lead}(s) = 0$ and the roots of $1 + G(s)G_{lead}(s)G_{lag} = 0$). For the closed-loop lead-lag system, the closed-loop pole near the origin is not the dominant pole. Why not?

Appendix A

Introduction to MATLAB and Control Systems Toolbox

A.1 Introduction

MatLab (The Mathworks Inc.) is a commercial "**Matrix Laboratory**" package which operates as an interactive programming environment for scientific and engineering calculations.

There are many different toolboxes available which extend the basic functions of Matlab into different application areas; in these exercises, we will make extensive use of the Control Systems Toolbox. Matlab is supported on Unix, Macintosh, and Windows environments. Most of the statements, functions and commands are computer platform independent. Regardless of what particular computer is used, the interaction with Matlab is basically the same.

Matlab is a command-driven, interactive language, aimed at the solution of problems involving vectors and matrices. The only data structure which Matlab uses is a non-dimensional matrix (or array), the dimensions being adjusted automatically by Matlab as required.

A.2 Statements and variables

Statements have the form:

```
>> variable = expression
```

Equals "=" implies the assignment of the expression to the variable. The command prompt is two right arrows ">>".

The assignment of value 1 to the variable a is executed after the enter key is pressed.

```
>> a = 1
a =
    1
```

The value of the variable is automatically displayed after the statement is executed. If the statement is followed by a semicolon (;) the output is suppressed.

```
>> a = 1;
```

The usual mathematical operators can be used in expressions. The common operators are "+" (addition), "-" (subtraction), "\" (division), "*" (multiplication), "^" (power). The order of arithmetic operations can be altered by using parentheses.

Matlab can be used in "calculator mode". when the variable name and "=" are omitted from an expression, the result is assigned to the generic variable *ans*.

```
>> 3.7*3
ans =
    11.1000
```

A.3 Entering vectors and matrices

A vector can be created by entering each element (separated by space) between brackets.

```
>> a = [1 2 3 4 5 6 9 8 7]
```

Matlab should return:

```
a =
    1    2    3    4    5    6    9    8    7
```

If you want to create a vector with elements between 0 and 20 evenly spaced in increments of 2 (this method is frequently used to create a time vector):

```
>> t = 0:2:20
t =
    0    2    4    6    8   10   12   14   16   18   20
```

You can reference individual items within the vector. To change the fifth element in the t vector:

```
>> t(5) = 23
t =
    0    2    4    6   23   10   12   14   16   18   20
```

Manipulating vectors is almost as easy as creating them. First, suppose you would like to add 2 to each of the elements in vector 'a':

```
>> b = a + 2
b =
    3    4    5    6    7    8   11   10    9
```

Now suppose, you would like to add two vectors together. If the two vectors are the same length simply add the two as shown below:

```
>> c = a + b
c =
    4    6    8   10   12   14   20   18   16
```

Subtraction of vectors of the same length works exactly the same way.

Entering matrices into Matlab is the same as entering a vector, except each row of elements is separated by a semicolon (;) or a return:

```
>> B = [1 2 3 4;5 6 7 8;9 10 11 12]
B =
     1     2     3     4
     5     6     7     8
     9    10    11    12
```

```
>> B = [ 1  2  3  4
        5  6  7  8
        9 10 11 12]
B =
     1     2     3     4
     5     6     7     8
     9    10    11    12
```

Matrices in Matlab can be manipulated in many ways. For one, you can find the transpose of a matrix using the apostrophe key:

```
>> C = B'
C =
     1     5     9
     2     6    10
     3     7    11
     4     8    12
```

It should be noted that if C had been complex, the apostrophe would have actually given the complex conjugate transpose. To get the transpose, use .' (the two commands are the same if the matrix is not complex). Now you can multiply the two matrices B and C together. Remember that order matters when multiplying matrices.

```
>> D = B * C
D =
    30    70   110
    70   174   278
   110   278   446
>> D = C * B
D =
   107   122   137   152
```

122	140	158	176
137	158	179	200
152	176	200	224

Another option for matrix manipulation is that you can multiply the corresponding elements of two matrices using the `.*` operator (the matrices must be the same size to do this).

```
>> E = [1 2;3 4], F = [2 3;4 5], G = E .* F
E =
     1     2
     3     4
F =
     2     3
     4     5
G =
     2     6
    12    20
```

If you have a square matrix, like E, you can also multiply it by itself as many times as you like by raising it to a given power.

```
>> E^3
ans =
    37    54
    81   118
```

If wanted to cube each element in the matrix, just use the element-by-element cubing.

```
>> E.^3
ans =
     1     8
    27    64
```

A.4 Functions

Matlab includes many standard functions. Each function is a block of code that accomplishes a specific task. Commonly used constants such as pi, and i or j for the square root of -1, are also incorporated into Matlab.

Matlab has available most trigonometric and elementary math functions of a common scientific calculator as shown in Table A.1.

Some functions for matrix properties and manipulation are given in Table A.2. To determine the usage of any function, type *help function name* at the Matlab command window.

Table A.1: Trigonometric and elementary math functions

<code>sin(x)</code>	Sine of the elements of x
<code>cos(x)</code>	Cosine of the elements of x
<code>asin(x)</code>	Arcsine of the elements of x
<code>acos(x)</code>	Arccosine of the elements of x
<code>tan(x)</code>	Tangent of the elements of x
<code>atan(x)</code>	Arctangent of the elements of x
<code>abs(x)</code>	Absolute value of the elements of x
<code>sqrt(x)</code>	Square root of x
<code>imag(x)</code>	Imaginary part of x
<code>real(x)</code>	Real part of x
<code>conj(x)</code>	Complex conjugate of x
<code>log(x)</code>	Natural logarithm of the elements of x
<code>log10(x)</code>	Logarithm base 10 of the elements of x
<code>exp(x)</code>	Exponential of the elements of x

Table A.2: Matrix manipulation

<code>inv(x)</code>	Inverse of a matrix x
<code>eig(x)</code>	Eigenvalues of the matrix x
<code>det(x)</code>	Determinant of matrix x
<code>rank(x)</code>	Rank of matrix x
<code>eye, ones, zeros, diag</code>	Array building functions

A.5 Polynomials

A polynomial is represented by a vector. To create a polynomial in Matlab, enter each coefficient of the polynomial into the vector in descending order. For instance, to enter the following polynomial:

$$p(s) = s^4 + 3s^3 - 15s^2 - 2s + 9$$

enter it as a vector in the following manner:

```
>> p = [1 3 -15 -2 9]
```

Matlab can interpret a vector of length $n+1$ as an n -th order polynomial. Thus, if your polynomial is missing any coefficients, you must enter zeros in the appropriate place in the vector. For example,

$$p(s) = s^4 + 1$$

would be represented in Matlab as:

```
>> p = [1 0 0 0 1];
```

Some functions to be used for polynomials are given in Table A.3:

Table A.3: Functions for polynomials

roots(p)	Roots of polynomial p
polyval(p,value)	Value of polynomial p at value
conv(p,q)	Polynomial multiplication
deconv(p,q)	Divide two polynomials

A.6 Loops and logical statements

Matlab provides loops and logical statements for programming, like *for*, *while*, and *if* statements. The general forms are:

for variable = expr, statement, ..., statement *end*;

while variable, statement, ..., statement, *end*

if variable, statements, *end*

A.7 Plotting

The simple *plot(x,y)* function will plot the vector x versus vector y. In Table A.4 there are several plot formats available in Matlab:

Table A.4: Plotting

plot(x,y)	Plots vector x versus vector y
semilogx(x,y)	Plots vector x versus vector y. The x-axis is log10, the y-axis is linear
semilogy(x,y)	Plots vector x versus vector y. The y-axis is log10, the x-axis is linear
loglogx,y)	Plots vector x versus vector y. The x-axis is log10, the y-axis is log10
mesh(x,y,z,c)	Creates a 3-D mesh surface

Suppose you wanted to plot a sine wave as a function of time. First make a time vector and then compute the sin value at each time.

```
>> t=0:0.25:7; y = sin(t); plot(t,y)
```

Basic plotting is very easy in Matlab, and the plot command has extensive add-on capabilities. These capabilities include many functions such as

A.8 Toolboxes and m-files

There are many functions in Matlab of direct use in control engineering. The Control Systems Toolbox was written to make use of these commands which extend Matlab's basic facilities. This provides single commands for such things as Bode plots, time responses, state-variable feedback and so on. There are many other toolboxes available for MATLAB,

Table A.5: Functions for plotting

grid	Toggles a grid on and off in the current figure.
axis	Controls axis scaling and appearance
line	Creates a line in the current figure
title('text')	Adds 'text' at the top of the current axis
xlabel('text')	Labels the x-axis with 'text'
ylabel('text')	Labels the y-axis with 'text'
subplot	Creates axes in tiled positions

e.g: Optimization, Identification, Image Processing, Neural Networks, Spline Functions, Robust Control, Adaptive Control, Symbolic Math etc.

The way these toolboxes work is of some relevance. They are actually written in MATLAB (that is, they use the statements and commands of the MATLAB language). They consist of collections of files, called m-files (because they have the filename extension .m). Anyone can write a m-file. It is just an ASCII file created using any ASCII text editor, and containing a sequence of MATLAB commands, typed exactly as they would be from the keyboard when using MATLAB interactively with nothing else added.

Example. Create an m-file called *garbage.m* containing nothing but the following lines:

```
a=[1 2 3; 2 84; 1 7 9];
inv(a)
eig(a)
```

and simply enter the filename (without the .m extension) in response to the MATLAB prompt. This will execute the commands in the file.

```
>> garbage
```

would have exactly the same result as entering the original commands. The file *garbage.m* has effectively become a new MATLAB command. It is often called a script file.

Another type of m-file is a **function** file. In contrast with the script files the function files has a name following the word "function" at the beginning of the file. The filename has to be the same as the function name The function statement syntax is:

```
function[output_arguments] = function_name(input_arguments)
```

The input arguments are variables passed to the function. The output arguments are returned.

Example. Create a m-file called *myfunc.m* which contains the following lines:

```
function [sum, product] = myfunc(x,y)
    sum = x+y;
    product = x*y;
```

The function will return the sum and product of two numbers and it can be called in the following way:

```
>> a=10;
>> b=25.9;
>> [alpha,beta]=myfunc(a,b)
or simply
>> [alpha,beta]=myfunc(10, 25.9)
```

The variable `alpha` will have the value of the sum of `a` and `b` and `beta` the value of the product.

Control System Toolbox

The Control System Toolbox builds on the foundation of MATLAB to provide specialized tools for control system engineering. The toolbox is a collection of algorithms, written primarily as M-files, that implement common control system design, analysis, and modeling techniques.

The Control System Toolbox is a core toolbox for the analysis, design, and tuning of feedback control systems. Its broad range of capabilities encompasses both classical and modern control design methods, including root locus, pole placement, and LQG regulator design. Convenient graphical user interfaces (GUIs) simplify typical control engineering tasks.

With the Control System Toolbox, you can model linear-time-invariant (LTI) systems in transfer function, zero/pole/gain, or state-space form. You can manipulate both continuous-time and discrete-time systems and convert between various model representations. You can compute and graph time responses, frequency responses, and root loci. Other functions let you perform pole placement, optimal control, and estimation. The Control System Toolbox is open and extensible, allowing you to create custom M-files to suit your particular application.

The most relevant functions are given below. For a complete description type *help function name*.

Model dynamics

pole, eig	System poles.
zero	System (transmission) zeros.
pzmap	Pole-zero map.
damp	Natural frequency and damping of system poles.

Time response

step	Step response.
impulse	Impulse response.
initial	Response of state-space system with given initial state.
lsim	Response to arbitrary inputs.

Frequency response

bode	Bode plot of the frequency response.
nyquist	Nyquist plot.
margin	Gain and phase margins.

System interconnections

parallel	Generalized parallel connection
series	Generalized series connection
feedback	Feedback connection of two systems

Classical design tools

rltool	Root locus design GUI
rlocus	Evans root locus
rlocfind	Interactive root locus gain determination

Demonstrations

ctrldemo	Introduction to the Control System Toolbox.
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Appendix B

Mathematical background

B.1 Linear approximation

The Taylor series expansion about the operating point $x_0=(x_{10}, x_{20}, \dots, x_{n0})$ is useful for a linear approximation to a nonlinear function. When the higher-order terms are neglected, the linear approximation of a function

$$y = g(x_1, x_2, \dots, x_n)$$

is written as:

$$y = g(x_{10}, x_{20}, \dots, x_{n0}) + \left[\frac{\partial g}{\partial x_1} \right]_{x=x_0} (x_1 - x_{10}) + \left[\frac{\partial g}{\partial x_2} \right]_{x=x_0} (x_2 - x_{20}) + \dots + \left[\frac{\partial g}{\partial x_n} \right]_{x=x_0} (x_n - x_{n0})$$

B.2 Laplace transform

1	Transform integral	$f(t)$	$\int_0^\infty f(t)e^{-st}dt$ or $\mathcal{L}[f(t)] = F(s)$
2	Linearity	$f_1(t) \pm f_2(t)$	$F_1(s) \pm F_2(s)$
3	Constant multiplication	$af(t)$	$aF(s)$
4	Complex shift theorem	$e^{\pm at}f(t)$	$F(s \pm a)$
5	Real shift theorem	$f(t-T)$	$e^{-Ts}F(s), T \geq 0$
6	Scaling theorem	$f(\frac{t}{a})$	$aF(as)$

7	First derivative	$\frac{d}{dt}f(t)$	$sF(s) - f(0)$
8	n-th derivative	$\frac{d^n}{dt^n}f(t)$	$s^n F(s) - \sum_{r=1}^n \frac{d^{r-1}}{dt^{r-1}}f(0)s^{n-r}$
9	First integral	$\int_0^t f(t)dt$	$\frac{1}{s}F(s)$

Table B.2: Laplace transform pairs

$f(t)$	$F(s)$
$\delta(t)$, unit impulse	1
1, unit step	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{-at}	$\frac{1}{s+a}$
$1 - e^{-at}$	$\frac{a}{s(s+a)}$
$\sin\omega t$	$\frac{\omega}{s^2+\omega^2}$
$\cos\omega t$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin\omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos\omega t$	$\frac{(s+a)}{(s+a)^2+\omega^2}$

Appendix C

Analysis of first- and second-order systems

C.1 Step response of a first-order system

Consider a first-order system with input $r(t)$ and output $c(t)$ as shown in Figure C.1. The

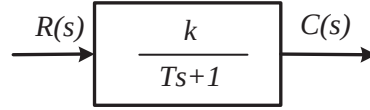


Figure C.1: First-order system

input is a unit step

$$r(t) = 1, \quad t \geq 0; \quad R(s) = \frac{1}{s}$$

The output $c(t)$ is:

$$\begin{aligned} c(t) &= \mathcal{L}^{-1}[C(s)] = \mathcal{L}^{-1}\left[\frac{1}{s} \cdot \frac{k}{Ts+1}\right] = \\ &= \mathcal{L}^{-1}\left[\frac{k}{s} - \frac{kT}{Ts+1}\right] = k - k \cdot e^{-\frac{t}{T}} \end{aligned}$$

The steady-state value of the output is:

$$c(\infty) = \lim_{t \rightarrow \infty} c(t) = \lim_{t \rightarrow \infty} \left(k - k \cdot e^{-\frac{t}{T}}\right) = k$$

The settling time (i.e. the time when the system response attains 0.98 of its steady-state value) is:

$$t_s = 4T.$$

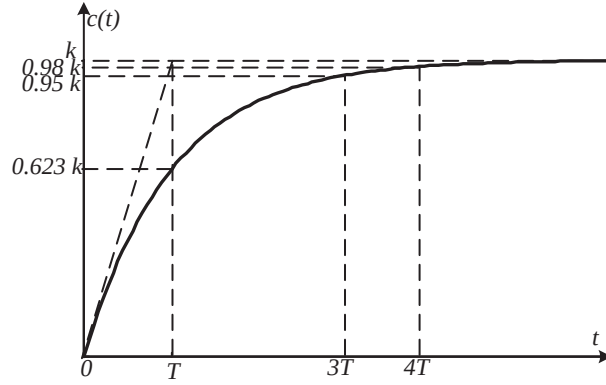


Figure C.2: Step response of first-order systems

C.2 Step response of a second-order system

Consider a second-order system with input $r(t)$ and output $c(t)$ as shown in Figure C.3. The input is a unit step

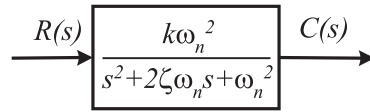


Figure C.3: Second-order system

$$r(t) = 1, t \geq 0; \quad R(s) = \frac{1}{s}$$

The output $c(t)$ is:

$$c(t) = \mathcal{L}^{-1}[C(s)] = \mathcal{L}^{-1} \left[\frac{1}{s} \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right]$$

1. $0 < \zeta < 1$: Underdamped case The location of the complex poles $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$ is shown in Figure C.4 where $\cos\beta = \zeta$ and $\sigma = \zeta\omega_n$.

The system response for a unit step input is shown in Figure C.5 and is given by:

$$c(t) = k \left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin \left(\omega_d t + \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta} \right) \right]$$

where $\omega_d = \omega_n\sqrt{1-\zeta^2}$ is the *damped natural frequency*.

Steady-state value:

$$c(\infty) = k$$

Maximum percent overshoot:

$$M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100 = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100$$

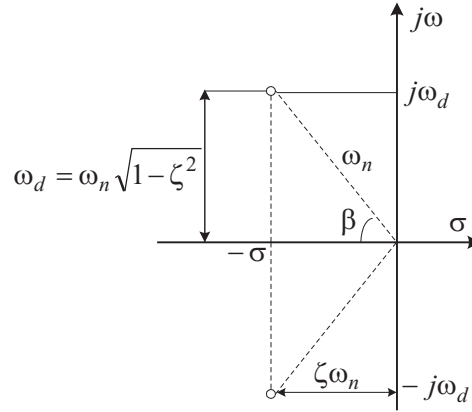


Figure C.4: Complex poles of a second-order system

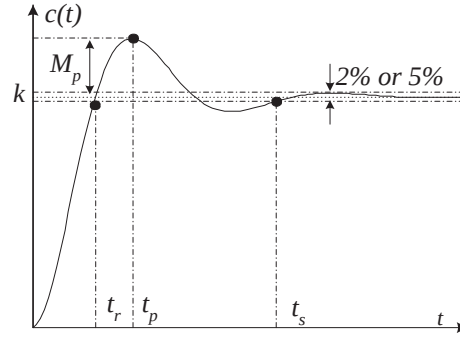


Figure C.5: Step response of a second-order system

Peak time:

$$t_p = \frac{\pi}{\omega_d}$$

Settling time (for the 2% criterion):

$$t_s = \frac{4}{\zeta \omega_n}$$

Rise time:

$$t_r = \frac{\pi - \beta}{\omega_d}$$

2. $\zeta = 1$: Critically damped case

$$c(t) = k \left[1 - e^{-\omega_n t} (1 + \omega_d t) \right]$$

3. $\zeta > 1$: Overdamped case

$$c(t) = k \left[1 + \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} \left(\frac{e^{-s_1 t}}{s_1} - \frac{e^{-s_2 t}}{s_2} \right) \right]$$

$$\text{where } s_{1,2} = \omega_n (\zeta \pm \sqrt{\zeta^2 - 1})$$

4. $\zeta < 0$: Unstable system

Appendix D

Root Locus

The **root locus** is the locus of roots of the characteristic equation as the gain is varied from zero to infinity.

For a typical closed-loop control system as the one shown in Figure D.1, the characteristic equation is:

$$1 + kG(s)H(s) = 0.$$

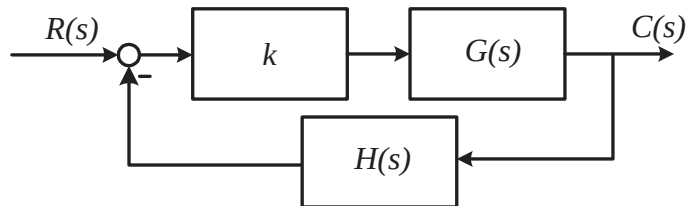


Figure D.1: Closed-loop system

Because s is a complex variable, the characteristic equation may be rewritten in polar form as:

$$|kG(s)H(s)| \angle kG(s)H(s) = -1 + j0$$

and therefore it is necessary that:

$$|kG(s)H(s)| = 1$$

$$\angle kG(s)H(s) = \pm 180^\circ(2k + 1), \quad k = 0, 1, 2, \dots$$

Root locus procedure

1. Write the characteristic equation so that the parameter of interest k appears as a multiplier

$$1 + kP(s) = 0.$$

2. Factor $P(s)$ in terms of n_p poles and n_z zeros

$$1 + k \frac{\prod_{i=1}^{n_z} (s + z_i)}{\prod_{j=1}^{n_p} (s + p_j)} = 0$$

3. Locate the open-loop poles and zeros of $P(s)$ in the s-plane with symbols: $\times$ - poles, $\circ$ - zeros.
4. Determine the number of separate loci SL . $SL = n_p$, when $n_p \geq n_z$, n_p = number of finite poles, n_z = number of finite zeros.
5. Locate the segments of the real axis that are root loci:
- (a) Locus begins at a pole and ends at a zero (or infinity)
 - (b) Locus lies to the left of an odd number of poles and zeros
6. The root loci are symmetrical with respect to the horizontal real axis.
7. The loci proceed to the zeros at infinity along asymptotes centered at σ_A and with angles Φ_A .

$$\sigma_A = \frac{\sum(p_j) - \sum(z_i)}{n_p - n_z}$$

$$\Phi_A = \frac{2q + 1}{n_p - n_z} \cdot 180^\circ, \quad q = 0, 1, 2, \dots, (n_p - n_z - 1)$$

8. By utilizing the Routh-Hurwitz criterion, determine the point at which the locus crosses the imaginary axis (if it does so).
9. Determine the breakaway point on the real axis (if any)
- (a) Set $k = -\frac{1}{P(s)} = p(s)$, (from the characteristic equation $1 + kP(s) = 0$)
 - (b) Obtain $dp(s)/ds = 0$
 - (c) Determine roots of (b) or use graphical method to find maximum of $p(s)$.
10. Determine the angle of locus departure from complex poles and the angle of locus arrival at complex zeros, using the phase criterion

$$\angle P(s) = \pm 180^\circ(2k + 1), \text{ at } s = p_j \text{ or } z_i.$$

11. Determine the root locations that satisfy the phase criterion

$$\angle P(s) = \pm 180^\circ(2k + 1) \text{ at a root location } s_x$$

12. Determine the parameter value k_x at a specific root s_x

$$k_x = \frac{\prod_{j=1}^{n_p} |s + p_j|}{\prod_{i=1}^{n_z} |s + z_i|} \Big|_{s=s_x}$$

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